



Deriving Vp velocity and density properties of complex litho-structural units from the analysis of geophysical log data: A study from the Southern Apennines of Italy

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ABSTRACT

This study is aimed to define a velocity and density database for the main litho-structural units of the Val d'Agri area of the Southern Apennines (Italy). Here we analyze fifteen digital borehole data comprising sonic log, check shot and density log data with dedicated graphic and statistical tools. We compare velocities acquired with two techniques: check-shots and sonic logging, discussing factors influencing them and their variations with a depth trend. In general, we observe that the comparison between the check shots and the sonic logs velocities gives consistent results. The detailed analysis of well logs and velocity distribution histograms, allow us to distinguish the following eight litho-structural units and their representative velocities and densities: Liguride tertiary flysches; Apennine Carbonate Platform, Lagonegro - Clastics, Lagonegro - Carbonates; Irpinia 1; Irpinia 2 and 3; Volturino foredeep turbidites; Apulian Carbonate Platform. The density values were calculated through the velocity density relationship using the Gardner equation, and subsequently compared with the measured density values. The comparison shows good correlations with the adopted Gardner coefficient of $a = 0.30$ and an exponent $m = 0.25$ for carbonatic succession, while a Gardner coefficient of $a = 0.25$ and an exponent $m = 0.28$ for clastic lithologies.

The obtained results represent a key input for the future velocity and gravity modellings and for the time to-depth conversion of seismic reflection profiles, describing the vertical and lateral lithological heterogeneities between carbonate platforms, deep-sea basinal units and clastic/turbidite successions in the complex tectonic settings of the Southern Apennines.

The approach applied in this work, based on a shared analytical and graphical analysis of borehole data, can be readily extended to other case studies worldwide, being the analyzed formations lithologically comparable to many other fold-and-thrust belt rocks.

1. Introduction

Time-domain seismic reflection has been the most successful seismic method for identifying subsurface geologic conditions favorable to the accumulation of oil and gas. Consequently, velocity models represent the base for reliable geological interpretation, being crucial for a solid depth conversion of the seismic reflection data. Thus, accurate knowledge of the velocity of the rocks is fundamental obtaining an accurate seismic image of the subsurface (AlAli and Anifowose, 2021). However, velocities can be very difficult to be properly characterized, especially when the geological setting history results in a complex architecture of the involved geological formations. The Val d'Agri area, located in the axial region of the Southern Apennines, is characterized by complex stack of tectonic units and hosts the largest onshore oil and gas field in

Europe (Holton, 1999). Discovered in 1988, the hydrocarbon field has been intensely investigated, through various geophysical acquisitions, including 2D and 3D seismic surveys, and by drilling more than 80 wells. The abundance of subsurface data provides a unique opportunity of defining detailed geological and geophysical subsurface models, accounting for the stratigraphic and structural complexity of this portion of the Southern Apennines fold-and-thrust belt.

Starting from this peculiar framework, this paper is aimed to define an updated, detailed model of the P-wave velocity and density of the upper crust of the Val d'Agri area, constituted of a complex pattern of stacked units of platform carbonates, pelagic basin carbonates, flysch units and foredeep turbidite units.

The study is based on the analysis of a large geophysical dataset, including sonic logs, check-shots, density logs, gamma-ray logs and

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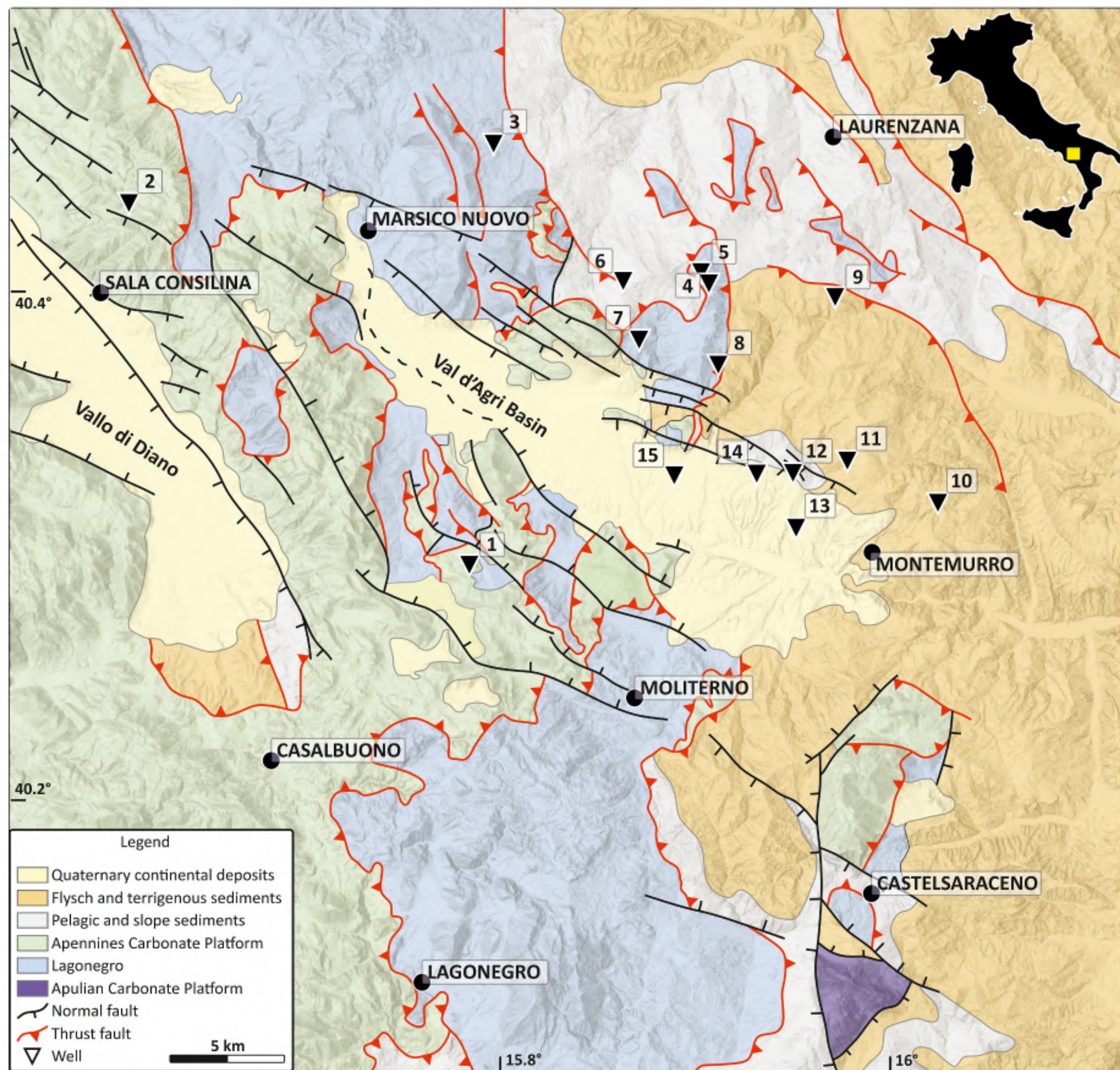


Fig. 1. Location map of the study area. The black triangles indicate the position of the wells analyzed in this study.

stratigraphic well tops, extracted from 15 deep boreholes located in a relatively restricted area (Fig. 1). The analysis of this complex subsurface dataset allows a complete characterization of the P-wave velocity and density of eight, relatively homogeneous litho-structural units, forming the subsurface setting of the study area.

While doing so, we define an original workflow, that can be easily extended to other structurally complex regions worldwide. In addition, since the litho-structural units considered in this study are similar in many other geological frameworks, particularly in fold-and-thrust belts, our results significantly contribute to the improvement of the geophysical databases and characterization of sedimentary rocks.

2. Geological setting

The Southern Apennines of Italy are a NW-SE trending Neogene - Quaternary foreland fold-and-thrust belt, formed in the complex framework of the Africa-Eurasia convergence and overlying the SW-dipping Apulian-Adriatic continental lithosphere (e.g., Scrocca, 2010 and references therein). The NE-verging fold-and-thrust belt and the associated foreland basins migrated eastward through time, incorporating a complex assemblage of tectonic units involving Mesozoic-Cenozoic successions, mostly deposited on the Adriatic

continental passive margin in carbonate platform or pelagic basin environment (D'Argenio et al., 1975; Mostardini and Merlini, 1986; Dewey et al., 1989; Patacca et al., 1989; Dercourt et al., 1992; Mazzoli et al., 2000, 2001).

Since the Late Miocene, the emplacement of the compressional belt to the east was synchronous with the co-axial extension at its rear, driven by the opening of the Tyrrhenian basin and progressively affecting the internal (western) and axial zone of the mountain belt.

The NE-SW trending Quaternary extension (Hippolyte et al., 1994) was accommodated by a series of NW-trending oblique-slip and normal fault systems of the Pliocene-Quaternary age (Brozzetti, 2011), cross-cutting the pre-existing contractional structures and leading to the opening of a set of intermountain basins, including the Val d'Agri basin (Cinque et al., 1993; Ferranti et al., 2005), located in the axial sector of the Apennine chain and filled by syntectonic continental Pleistocene deposits (Ascione et al., 1992; Cello et al., 2000; Giano et al., 2000).

This area of the Southern Apennines is affected by destructive earthquakes, with maximum magnitude up to 7.0 Mw, which reflects the still on-going, WSW-ENE trending extension (Amato and Montone, 1997; Hunstad et al., 2003; Burrato and Valensise, 2008; D'Agostino et al., 2008; Montone and Mariucci, 2016).

The subsurface setting of the Val d'Agri area consists of a complex

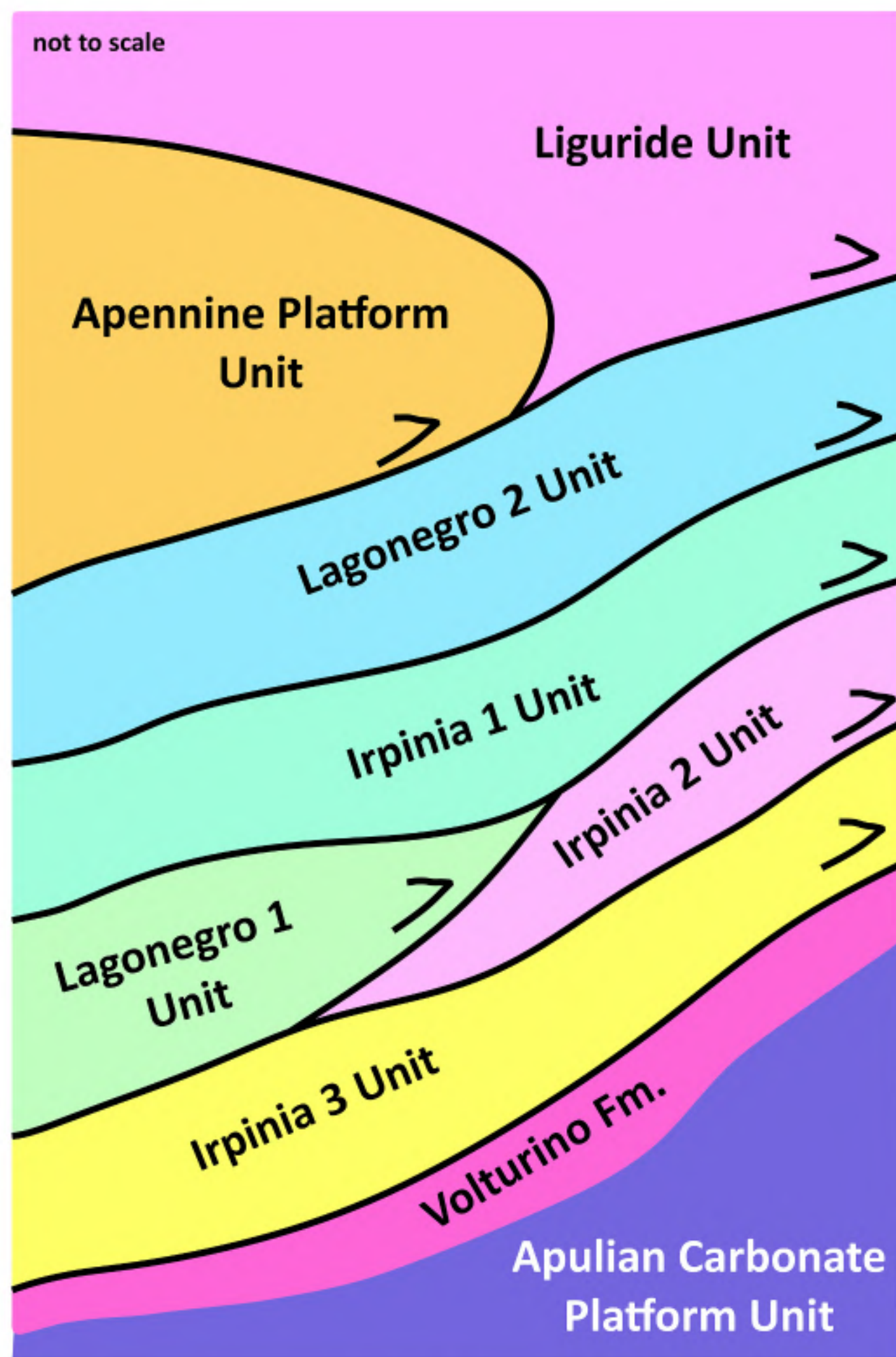


Fig. 2. Schematic representation of the geometric relationships between the tectonic units of the Val d'Agri area (modified after Pajola et al., 2020).

stack of tectonic units, schematically represented in Fig. 2 and briefly described in the following, top to bottom.

- The Liguride Tertiary flysch Unit, originally deposited in the internal domains associated with the Ligurian-Piedmont paleo-ocean; and the Apennine Carbonate Platform, the Lagonegro Basinal and the Apulian Carbonate Platform Units, all originally deposited on the continental passive margin of the Adria Plate: The Liguride Unit (Lower Cretaceous - Lower Miocene) includes a Tertiary flysch succession, mainly composed of mixed deep-sea siltstones and claystones, overlying the ophiolite-bearing sediments derived from the Ligurian-Piedmont branch of the Neo-Tethyan Ocean (e.g., Ogniben, 1969; Knott, 1987; Bonardi et al., 1988; Monaco and Tortorici, 1995; Lentini et al., 2002; Cavalcante et al., 2012).
- The Apennine Carbonate Platform Unit (Upper Triassic - Lower Miocene) consists of an up to 5000 m thick succession of shallow-water carbonates and transitional slope-to-basin resedimented limestones (D'Argenio et al., 1973; D'Argenio et al., 1975; Palladino et al., 2008; Mostardini e Merlini, 1986; Menardi Noguera e Rea, 2000; Van Hinsbergen et al., 2020), overthrusting the coeval Lagonegro Unit.
- The Lagonegro Basinal Units (Middle Triassic - Early Tertiary) consists of a deep-sea basin succession, which is internally stacked to form tectonic duplexes overthrusting the buried internal portion of the Apulian Carbonate Platform. At a regional scale, the Lagonegro units are constituted by two superimposed major thrust sheets: i) the

upper thrust sheet (Lagonegro 2), about 2000 m thick, composed of proximal marine deposits with abundant shallow-water carbonates; and ii) the lower thrust sheet (Lagonegro 1), about 1500 m thick, composed of deep-sea pelagic deposits with distal facies (e.g., Scandone, 1972; Menardi Noguera and Rea, 2000; Patacca and Scandone, 2007a,b). From a lithological point of view, the Lagonegro Units are a complex multilayer (e.g., Marsella et al., 1995), including siliceous shales, marls, siliciclastic calcarenites and subordinate fine-grained sandstones.

- The Irpinia Units include the Tertiary thrust-top succession, deposited in the inner margin of the Irpinian foredeep basin (Pugliese et al., 2012), and the thick allochthonous complex, overthrusting the lower Pliocene foredeep turbidites successions, deposited above the Apulian Carbonate Platform (Carubelli et al., 2010). The term "Irpinia units" is derived from the work of Menardi Noguera and Rea (2000), Carubelli et al. (2010) and Pajola et al. (2020), who subdivided it into three sub-units (Irpinia 1, 2 and 3), in order to constrain the time-space evolution of the thrust belt. As these units are tectonically complex, and interposed between Lagonegro and Apulian Carbonate Platform Units, they have also been identified as a "tectonic mélange" by Mazzoli et al. (2008, 2014).
- The Apulian Carbonate Platform Unit is composed of a very thick succession of Mesozoic- Early Tertiary shallow-water carbonates (Lentini et al., 2002), stratigraphically overlain by a relatively thin section of Miocene transgressive limestones and evaporites (Messinian mudstones, anhydrides, gypsum) and by a lower Pliocene foredeep turbidite sequence (Patacca and Scandone, 2007). The lower Pliocene succession, here informally named Volturino Fm. (Carubelli et al., 2010), consists of a monotonous sequence of medium to thick bedded, very coarse to medium graded sandstone (and pebbly mudstone), with erosive base and abundant shale clasts, also characterized by a relatively high content in carbonate lithic fragments (Carubelli et al., 2010).

3. Methods

3.1. Check shots and sonic logging acquisition techniques and related errors

In this work, we analyzed and compare borehole velocity data acquired with check-shots and sonic logging techniques, which are currently the most common tools used for determining borehole velocity at a given location.

Check-shot survey directly measures average velocities by recording transit times and depths at comparatively coarse intervals (61–305m) and lower frequencies (~50 Hz) from the surface to a series of geophones/receivers installed along the borehole. Usually, check-shot data are the most effective correlation tool available between the wellbore and the seismic reflection data, thus, they are used to derive the P-wave velocity of a specific layer of rock (the so-called interval velocity).

Check shot data is typically collected during the drilling of a well and provides information about the time it takes for seismic waves to travel through the subsurface formations. This data can then be used to calculate the Vp throughout the following steps:

- Check shot data collected by deploying a geophone at the surface.
- The source of seismic waves is triggered at known depths in the well.
- Measured the arrival times of seismic waves at the surface for each shot point. This data is recorded as a function of depth.
- Identified the first arrivals on the recorded seismogram for each shot point. These first arrivals are the direct P-waves that have travelled through the subsurface formations.
- Plotted the travel times of the P-waves as a function of depth. This provided a time-distance curve.
- Corrected the time-distance curve for borehole effects, elevation differences, and surface wave interference.

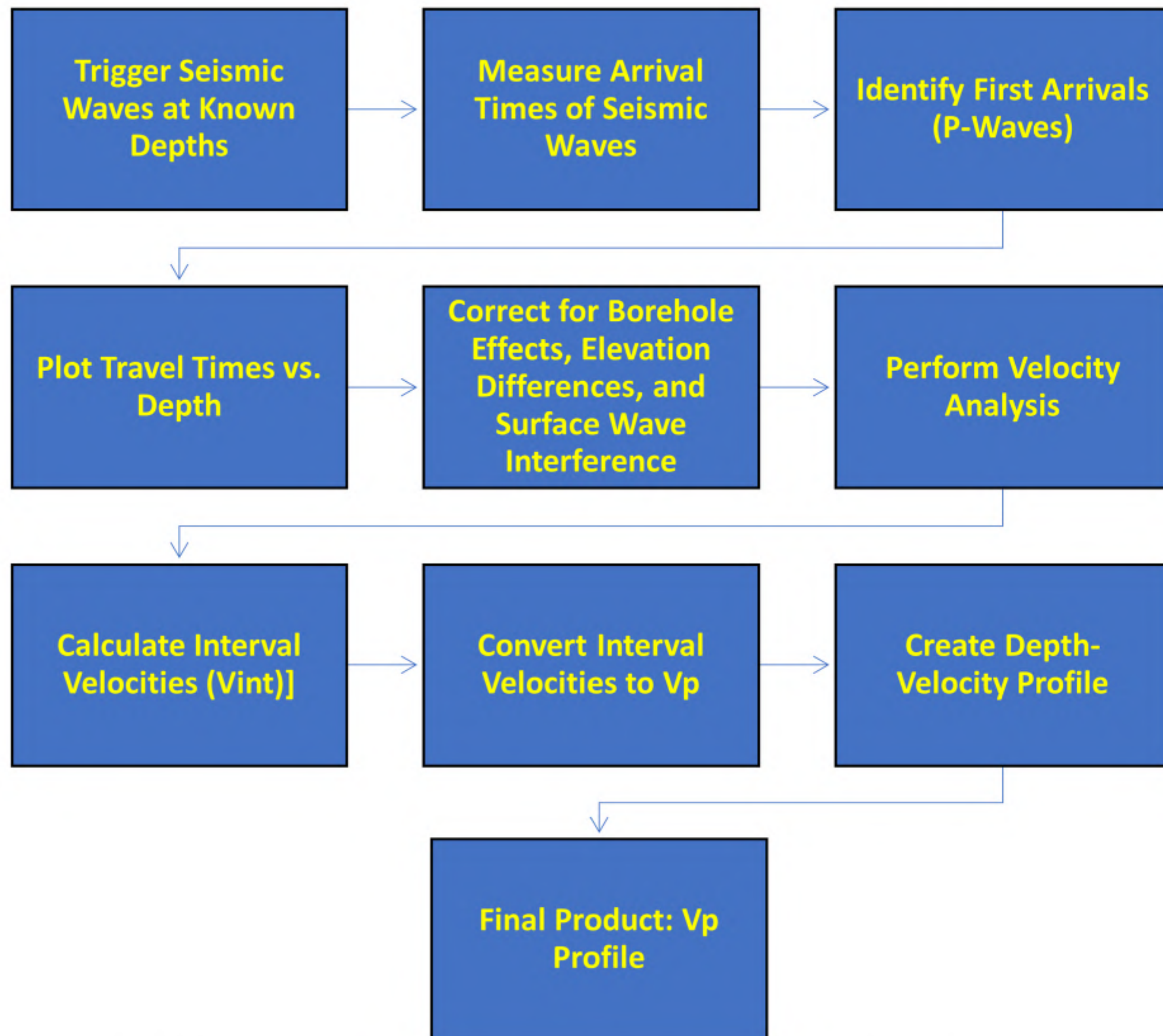


Fig. 3. The workflow visually outlines the sequential steps involved in obtaining the P-wave velocity (Vp) profile from check shot data collected during well drilling. Each step is connected to the next, and the final product is the Vp profile, which is crucial for subsequent geological and geophysical analyses.

- Performed velocity analysis to convert the travel times into interval velocities (Vint) for each depth level by applying the equation:

$$V_{int} = (\Delta T) / (2 * \Delta D)$$

Where:

- Vint - interval velocity.
- ΔT - travel time between two depth levels.
- ΔD - depth interval between those levels.

- Then converted Interval Velocities to Vp. To obtain the P-wave velocity (Vp) at each depth level, used the following relationship:

$$V_p = 1 / (V_{int})$$

- Created a depth-velocity profile that allowed to define how Vp varies with depth in the well. This final product for each well where check-shot was acquired is received and used for present study analysis (Fig. 3).

Conversely, the sonic logging tool provides a continuous record of interval transit time measured across comparatively much shorter intervals (typically of the order of ~ 0.6 m) and higher frequencies (around 20–25 kHz) with respect to the check-shot survey. Since the inverse of interval transit time is velocity, it also provides information regarding

both punctual and interval velocities when measures are grouped following the lithological variations.

It should be noted that velocity values inferred by sonic log recordings may differ from those obtained using check-shots (Goetz et al., 1979). The reasons behind such misfit lie on the different measurement principles. Therefore, in geophysical practice, sonic logs should be calibrated with check shots for several reasons. First, a sonic log typically starts several hundreds of meters below the datum: therefore, during the calibration process, sonic logs are verticalized (True Vertical Depth) and referenced to the seismic reference datum. Secondly, the sonic log measures the velocity in a frequency range much higher than the seismic frequency range, so the velocities may be biased by frequency due to the dispersion (e.g., Mavko et al., 2009). Moreover, sonic log recordings may contain some erroneous values due to borehole problems, such as alteration and/or invasion resulting in cycle skipping, tool sticking and downhole conditions such as washed-out zones in a borehole (Gretener, 1961; Goetz et al., 1979). Errors associated with sonic logs measurements must be considered, which is why sonic logs' editing is frequently required and usually performed during the sonic log calibration processes.

3.2. Data treatment

For all the considered fifteen wells (Fig. 1), we collected the available

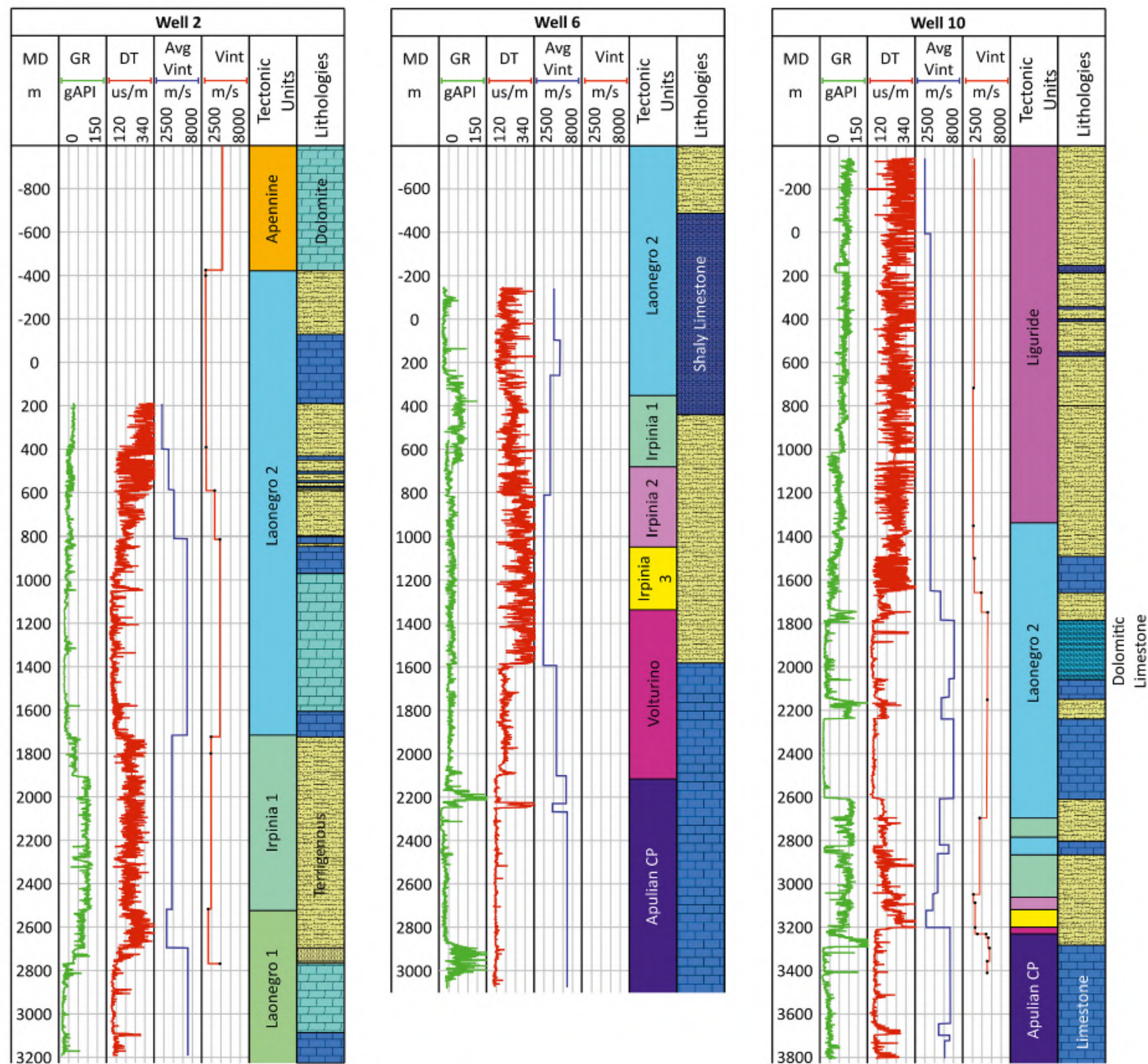


Fig. 4. Examples of analyzed wells. From left to right: Gamma-Ray log (green curve), Sonic log (red curve), light, Velocity logs (dark blue), Check-shots (discrete red curve), tectonostratigraphic units and lithologies.

log data, including gamma-ray (GR), sonic (DT) and density (RHOB) logs, along with check-shots and wells stratigraphy (provided in Fig. S1).

The calibrated sonic logs are converted from interval transit time (us/m) to velocity (m/s), and then velocities distribution in the main litho-stratigraphic units of the study area have been analyzed. Subsequently, to study the scale effect on the velocity measurements, the smaller-scale (higher frequency) sonic log velocities are compared directly with the larger-scale (lower frequency) check-shot interval velocities in each well where both data are available. To easily compare the two dataset values, the sonic logs are edited by the log conditioning process, i.e., by the automatic *blocking* process in the Petrel software (mark of Schlumberger). We generated new discrete logs (dark blue curve in Fig. 4), where an average constant velocity is associated with automatically defined depth intervals characterized by relatively homogeneous velocities. The resulting discrete velocity logs were compared with interpolated check-shots (discrete red curves in Fig. 4), available for the Well 11, Well 12, Well 7, Well 2 and Well 10.

Further, the extracted velocities were assigned to the tectonic units interpreted by ENI company stratigraphers and provided for our study, which are the Liguride, Lagonegro 1 and 2, Irpinia 1, 2, and 3 and Apulian Carbonate Platform. The heterogeneous distribution of the obtained interval velocities within the Lagonegro 1, Lagonegro 2 and

Apulian Carbonate Platform tectonic units have been interpreted as caused by the presence of different alternating lithologies. This is demonstrated by the detailed lithological description present in Well 2, Well 10 and Well 6. In all three wells, the drilled sequences show terrigenous interlayers within the three tectonic units (VIDEPI database). To extend the detailed lithological variation in the other wells, where no lithological description is available, we performed an integrated analysis of the GR and DT logs (Fig. 5): using these data, we internally subdivided the Lagonegro Units into their alternated carbonatic and terrigenous layers, and separated the Apulian Platform carbonates from the Volturino terrigenous unit at its top.

Once units with similar lithological characteristics were properly identified, we extracted velocities for each unit and performed statistical analysis.

3.3. Verification of Gardner's equation and derivation of an empirical equation for density estimation

Density is a fundamental physical property of rocks, and it is usually obtained in a laboratory by analyzing core samples or during wireline logging. Both methods are quite expensive and time-consuming; therefore, density values are not often available. Several empirical equations

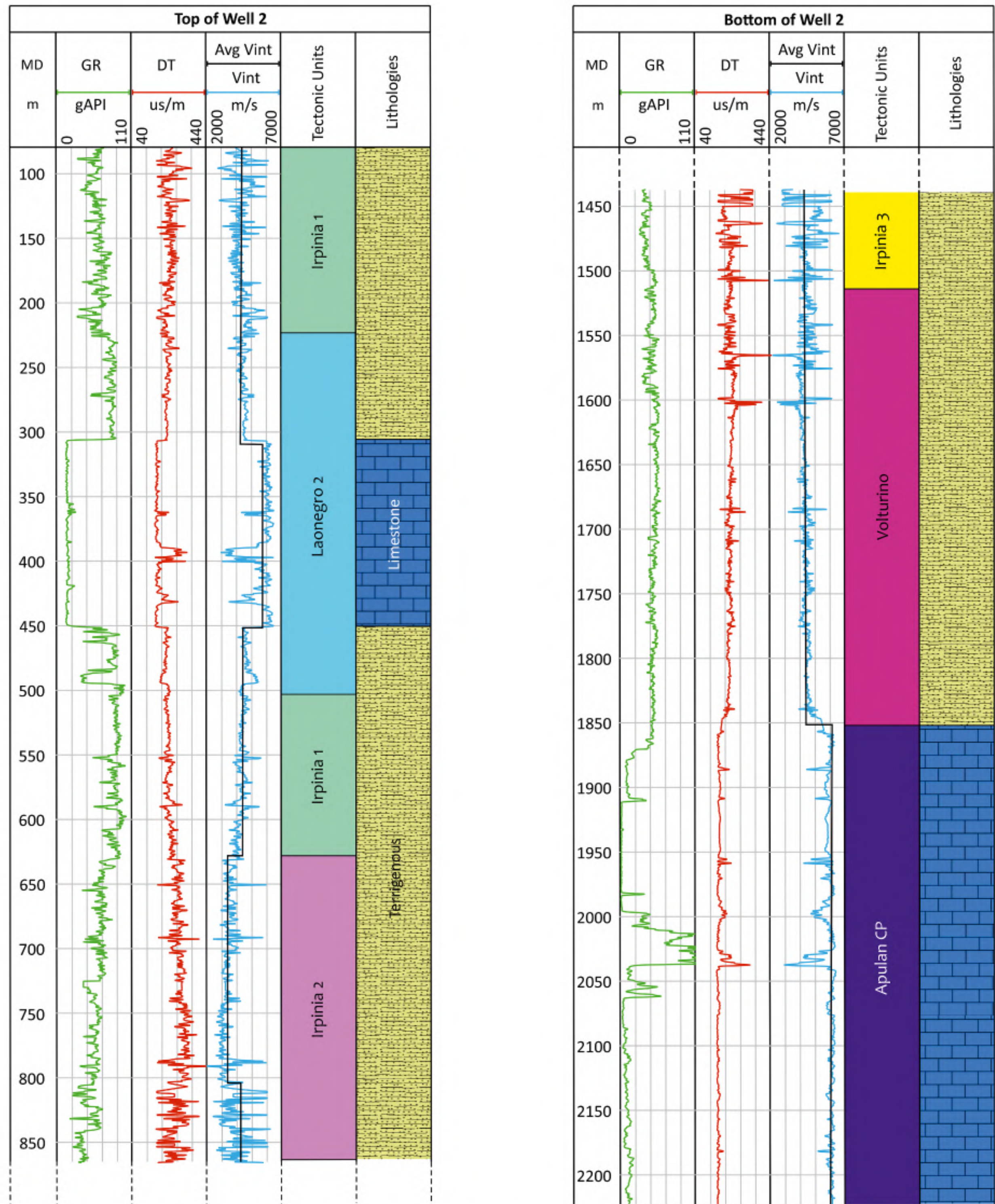


Fig. 5. Examples of the resulting discrete logs where carbonate and terrigenous layers are separated from each other within the Lagonegro 2 and Inner Apulian Platform units.

have been developed to infer velocity/density relationship (Castagna et al., 1985; Bowers, 1995). These equations consider several local factors such as local variations of lithology and fluid content. However, at larger scale, commonly, density is derived from synthetic density log, which in turn is calculated starting from velocity data. Gardner et al. (1974) found an empirical relationship between density and velocity from a series of controlled field and laboratory measurements of brine-saturated rocks, excluding evaporites, from various locations and depths, given by:

$$\rho = aV^m$$
 (Eq. 1)

where ρ is density expressed in g/cm^3 , and V is P-wave velocity, expressed in m/s . Default values suggested for a and m coefficient are 0.31 and 0.25, respectively. Gardner's relation is a good approximation for regional studies and for lithologies such as shales, sandstones, and carbonates, while coals and evaporites depart significantly from the expected behavior. Gardner's equation might not be appropriate for all regions; hence synthetic density calculated by the general equation might not give correct estimates because the deposition environment and the structural setting of the sedimentary basins vary from one region to another. Therefore, if field data is available, empirical data need to be

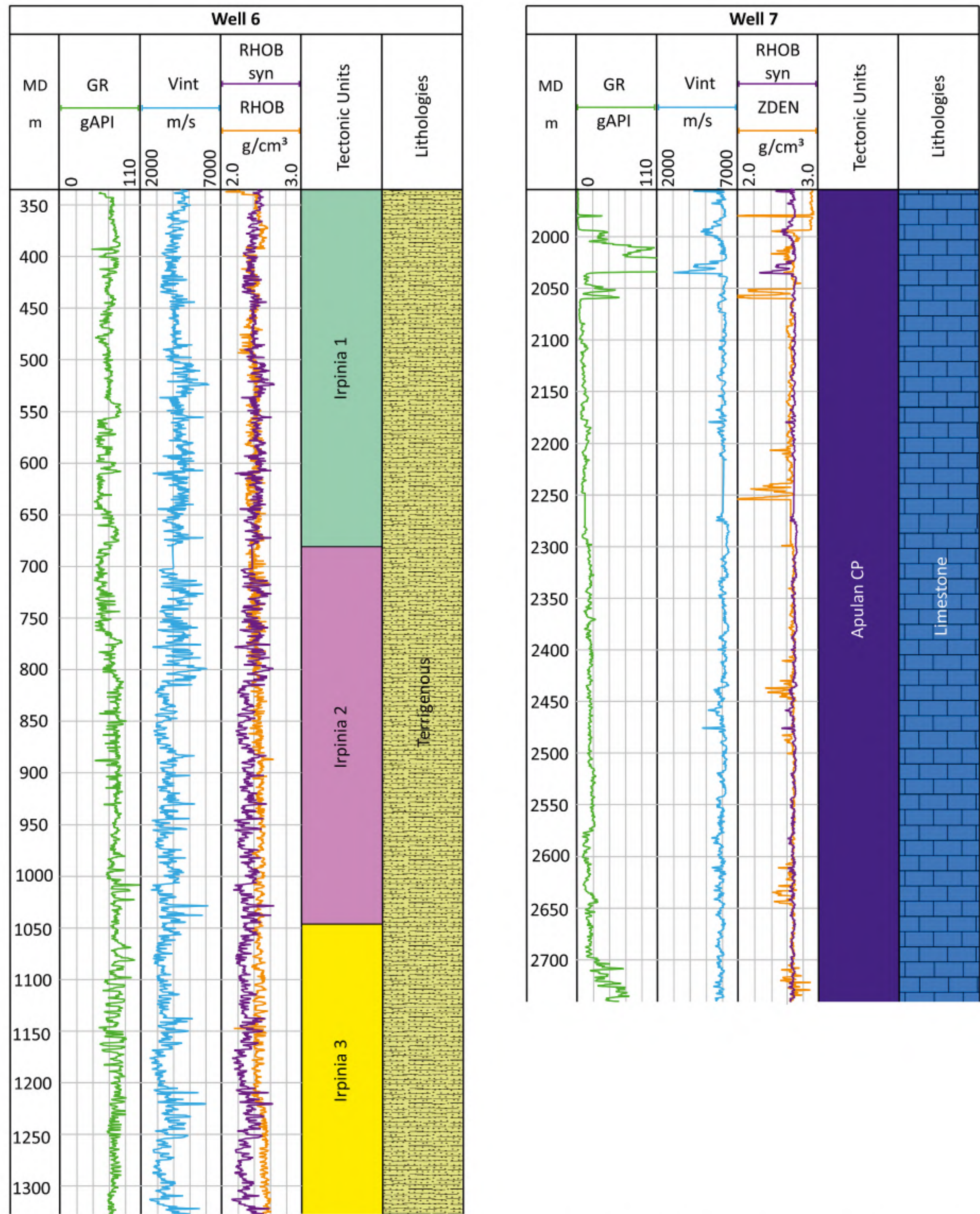


Fig. 6. Comparison of measured density log (orange curve) with synthetic density log (purple curve) obtained from Vp velocity log (light blue curve).

correlated to obtain better results (Gardner et al., 1974). For example, Ojha and Sain (2014), Gardner parameters were fitted using measured density and velocity values from well data (core and log) by plotting velocity against density and defining a power curve for the Kerala-Konkan basin of the western Indian margin. Gardner equation offer several advantages:

1. it is implemented into the Petrel software that we used for all our calculations and analysis.
2. it is a commonly used method for predicting density in the petroleum industry
3. it is easily comparable with most of the papers dealing with density/velocity relations paper.

The Gardner equation represents a simplified approach that may not capture all the nuances of local conditions, however it represent a perfect approach for a regional-scale study, consequently, following the main objectives of our study, we considered that on a regional scale, present minor uncertainties are acceptable.

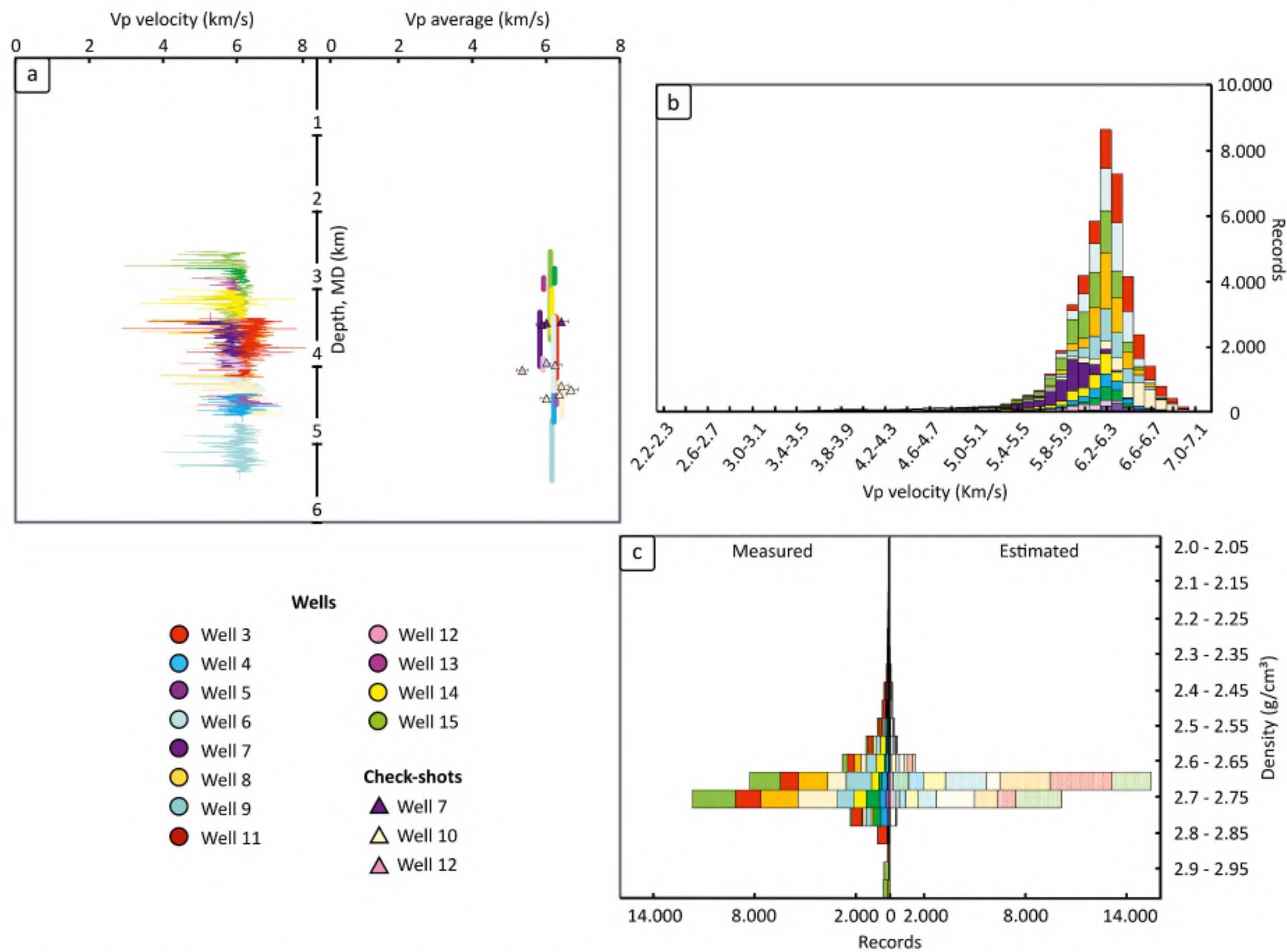


Fig. 7. Velocity and density analysis in the Liguride unit. (a) sonic logs and average Vp discrete logs vs depth cross-plot; (b) Vp distribution histogram; (c) the bar chart of the calculated density.

Here, out of the fifteen available wells, eleven have density logs, most of which measured only within the Apulian Carbonate Platform Unit. Thus, we applied Gardner's equation to these eleven wells, in order to verify its reliability for all the lithologies drilled by the wells. We used the velocities obtained from the calibrated sonic log, and the default values of a and m to calculate a synthetic density log (purple curve in Fig. 6), and visually compared it with existing measured density logs (orange curve in Fig. 6).

Following the comparison between the density calculated from synthetic logs and the available measurements of bulk density logs, we find that the used default Gardner parameters are reliable to indirectly obtain density values, when a density log was not acquired. However, following a detailed statistical analysis performed unit by unit, we observe discrepancies in the distribution of the two densities (i.e., calculated and measured). Thus, we re-run the density calculations after calibration, adopting $a = 0.30$ and $m = 0.25$ for carbonatic succession; and $a = 0.25$ and $m = 0.28$ for terrigenous lithologies.

4. Data analysis and results

This section analyzes the velocities derived from calibrated sonic logs and reports both the seismic velocities and densities of the eight considered litho-structural units, from top to bottom. We describe results in the following order: i) Vp from sonic logs versus depth cross plot; ii) Vp discrete logs and check-shots versus depth cross plot; iii) Vp distribution histogram; iv) bi-directional bar chart of the calculated synthetic density logs against measured densities. The penetrated depth and thickness of the units, together with the average velocities and the average depth for all wells, are compiled in Table S1 in the Supplementary material.

4.1. Apennine Carbonate Platform unit

Among the two wells penetrating this unit, only Well 2 has velocity data from the check-shot, indicating a velocity of 6000 m/s, while Well 1 lacks any velocity data. Therefore we did not perform any data analysis for this unit.

4.2. Liguride tertiary flysch unit

Among the four wells penetrating this unit, the sonic logs and discrete Vp logs show a slight velocity increase with depth (Fig. 7a). A similar trend is obtained for the interval velocities from check-shots (triangles on Fig. 7a).

The histograms of velocity distribution (Fig. 7b) show recorded velocities characterized by unimodal Gaussian distribution, with a mode at 3600–3800 m/s. The highest mean velocity value is 4024 m/s (Well 10), while the lowest is 3504 m/s (Well 13); the average velocity for the entire unit is 3763 m/s.

The shallowest occurrence of the unit (Well 11) is at 193 m MD, while the most profound (Well 10) is at 2334 m MD. The penetrated thickness ranges between 293 m (Well 12), and 1734 m (Well 10).

Due to the lack of measured density logs, Fig. 6d shows only the calculated density values for the Liguride unit. The histogram shows that most of the data ranges between 2.3 g/cm³ and 2.4 g/cm³, while the calculated average density is 2.36 g/cm³.

4.3. Lagonegro Units

Fifteen wells penetrated Lagonegro 1 and/or Lagonegro 2 pelagic basin units (Figs. 8 and 9). The maximum penetrated thickness is 1878 m (Well 5), while the minimum is 278m (Well 15). Wells 3, 15, 7 and 2

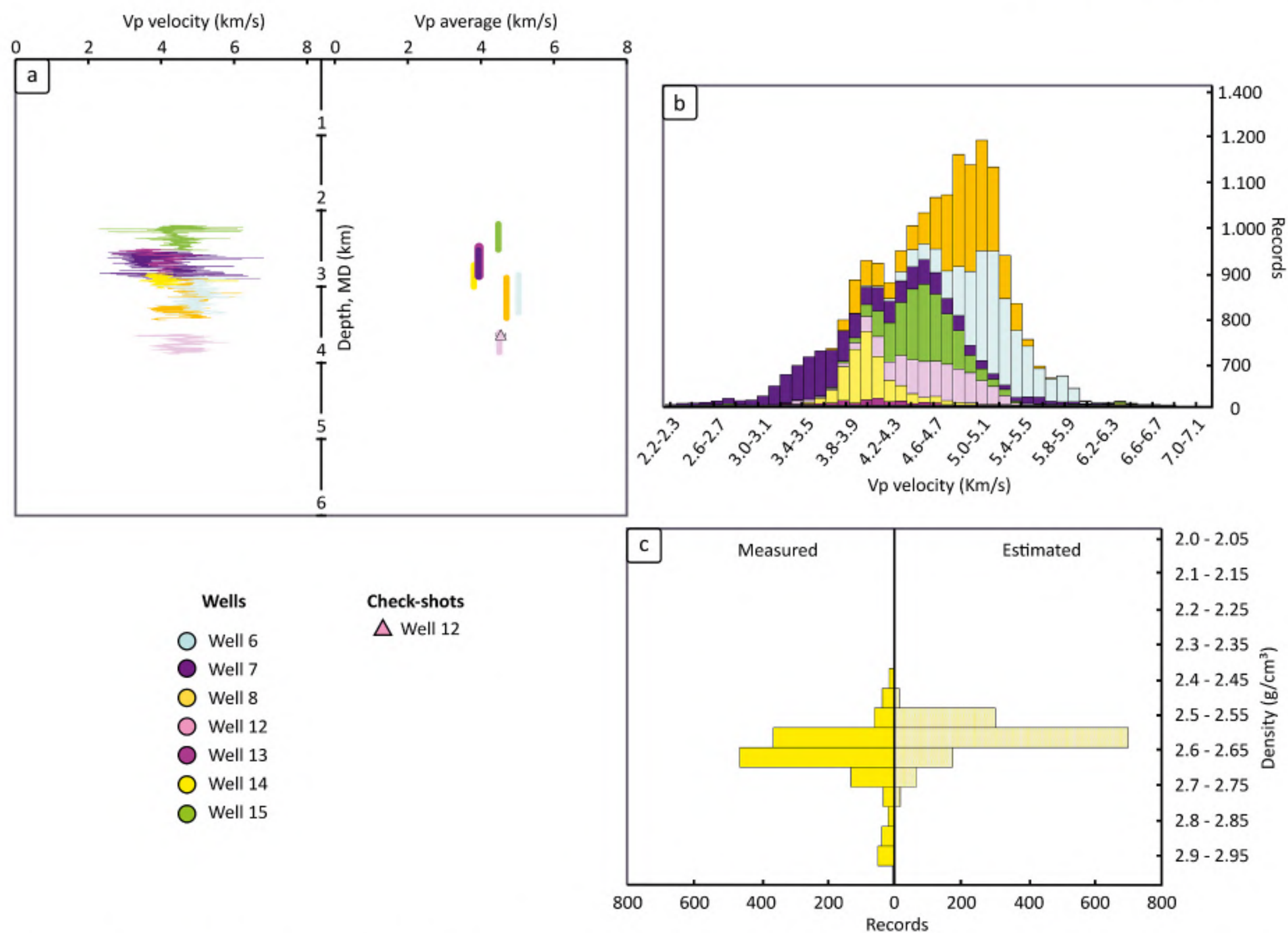


Fig. 8. Velocity and density analysis of the Lagonegro - Clastic units. (a) sonic logs and average Vp discrete logs vs depth cross-plots; (b) Vp distribution histogram (c) the bi-directional bar chart of the measured and calculated densities.

encountered two distinct thrust sheets of the Lagonegro 1 and 2 units, separated by Irpinia units.

Lagonegro 1 and Lagonegro 2 units show a wide velocity variation ranges, going from 4000 m/s to 6500 m/s, with two modes of velocity distributions. The detailed lithological description from the reference wells confirmed that Lagonegro units comprise both turbiditic clastics with lower velocities intercalated with pelagic carbonates with higher velocities. Consequently, we did not analyze the Lagonegro 1 and Lagonegro 2 units as separate units; instead, we performed an analysis based on lithology, i.e. predominantly terrigenous and predominantly carbonatic lithologies, due to their remarkable differences in velocity (Lagonegro-Clastic and Lagonegro-Carbonate in the following analyses).

To distinguish the carbonates from the clastic, we used the composite logs with detailed lithological descriptions from three wells (Well 2, Well 10 and Well 6), available in the VIDEPI database. These wells have been successfully used as the reference for the lithology discrimination in the other wells, where lithological logs are absent, and carbonates were discriminated from clastics by integrated interpretation of GR and DT logs. As a result, we obtained new discrete lithological logs that allowed us to separate velocities for the main lithology. This division comprises the separation of mainly gross intervals (thickness >100 m), characterized by prevalence of clastics vs. carbonates.

Lagonegro-Clastic represents units with predominantly clastic lithology encountered in twelve out of fifteen wells. The sonic logs and discrete Vp logs illustrated in Fig. 8a show a gentle velocity increase with depth, particularly at shallow depths. A similar trend is observed for the interval velocities from check-shots (triangles in Fig. 8a). The histogram of velocity distribution (Fig. 8b) shows a wide velocity range aligned into a cumulative gaussian unimodal distribution at 4300–4800 m/s, with some outlier values to the left. The highest mean velocity value is 4926 m/s (Well 9), while the lowest is 3919 m/s (Well 12), and

the average velocity is 4429 m/s.

The shallowest occurrence of Lagonegro-Clastic (Well 5) is at 256 m MD, while the deepest (Well 10) is at 3820 m MD. The minimum penetrated thickness is 177 m (Well 8), while the maximum is 1198 m (Well 2).

In the Lagonegro units, the measured density log is present only for Well 11, where we calculated density and plotted it against measured density values. The Lagonegro-Clastic unit does not show a good agreement between measured and calculated data (Fig. 8c): the measured densities show a widespread distribution and an average value of 2.28 g/cm³, while calculated densities show Gaussian unimodal distribution, with an average value of 2.46 g/cm³.

Lagonegro - Carbonate represents units with a prevalent pelagic/carbonatic lithology, encountered in the thirteen wells. The sonic logs and discrete Vp logs illustrated in Fig. 9a show a detectable velocity increase with depth. A similar trend characterizes the interval velocities from the check-shots (triangles in Fig. 9a).

The velocity distribution histograms (Fig. 9b) show a unimodal velocity distribution with a left skew. Most of the data ranges between 5400 and 6700 m/s. The highest mean velocity value is 6594 m/s (Well 10), while the lowest is 5248 m/s (Well 15); the average velocity is 5855 m/s.

The shallowest occurrence (Well 15) is at 450 m MD, while the deepest (Well 2) is at 4307 m MD. The penetrated thickness ranges between 395 m (Well 1) and 1643 m (Well 3).

We calculated density and plotted it against measured density values, available only from Well 11. The comparison (Fig. 9c) shows a fairly good agreement between the average measured density (2.66 g/cm³) and the average calculated density (2.62 g/cm³).

Considering that the comparison (Fig. 9c) is based only on measured density from one well, we decided to check the reliability of the equation

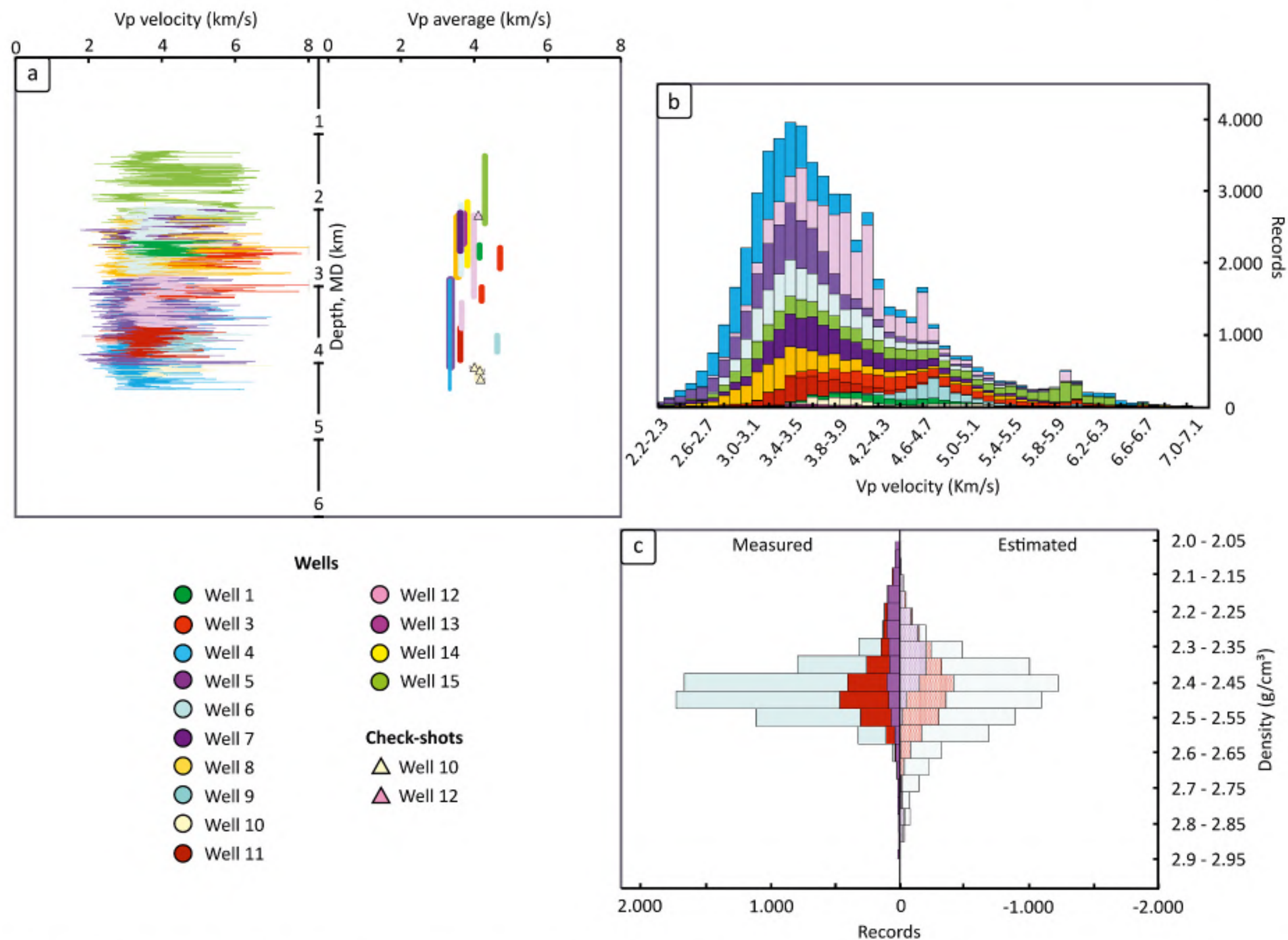


Fig. 9. Velocity and density analysis in the Lagonegro - Carbonate units. (a) sonic logs vs depth cross plot and average Vp discrete logs vs depth cross-plots; (b) Vp distribution histogram; (c) the bi-directional bar chart of the measured and calculated densities.

defined for clastics and carbonates by calculating the density for all studied wells where velocity logs are available.

- . The resulted average density is 2.4 g/cm³ (clastics) 2.63 g/cm³ (carbonates) (Fig. 10). Thus, we concluded that the defined coefficients for calculating synthetic density logs, could also be considered reliable in this case.

4.4. Irpinia 1 unit

Fourteen wells penetrated this unit. The cross-plot in Fig. 11a show a slight velocity increase with the depth. The check-shots records have a similar pattern of interval velocity in three wells except for Well 2, where interval velocities are lower than the sonic log velocity values (triangles in Fig. 11a).

The velocity distribution histogram is generally unimodal (Fig. 11b), where most of the data ranges between 4200 and 4900 m/s. The highest mean velocity is 5272 m/s (Well 9), while the lowest is 3716 m/s (Well 3); the average velocity is 4619 m/s.

The shallowest occurrence (Well 15) is at 732 m MD, while the deepest (Well 10) is at 4036 m MD. The total penetrated thickness ranges between of 80 m (Well 3) and 792 m (Well 2).

The measured density log is present only in Well 6. The comparison between measured and calculated density data (Fig. 11c) shows the same range of 2.40–2.45 g/cm³, with a similar average density of 2.40 g/cm³ and 2.42 g/cm³, respectively. Considering the fact that Well 6 has the lowest average velocity value among other analyzed wells, the calculated and measured densities might not be representative for the entire unit. Thus, we calculated density from sonic logs for all the wells (Fig. 12), obtaining a more representative average density of 2.47 g/cm³.

4.5. Irpinia 2 and Irpinia 3 units

Irpinia2 and Irpinia3 units, encountered in thirteen and fourteen wells, respectively, were analyzed together, since both lithology and velocities are similar. The sonic logs in Fig. 13a do not show any detachable increase with increasing depth. A similar pattern is also shown by the interval velocity derived from the check-shots (triangles in Fig. 13a).

The histogram in Fig. 13b shows a unimodal velocity distribution, where most of the data ranges between 3000 and 4100 m/s, with a left-skewed. The mean velocity values range between 4416 m/s (Well 3) and 3333 m/s (Well 4); the average velocity of the entire unit is 3880 m/s.

The shallowest occurrence of the unit (Well 15) is at 1227 m MD, while the deepest (well 9) is at 4699 m MD. The penetrated thickness ranges between 165 m (Well 10) and 1496 m (Well 12).

For the Irpinia 2 and Irpinia 3 units, measured density logs are present in three wells. The comparison between measured and calculated densities (Fig. 13c) shows a similar distribution range, with modes of 2.40–2.50 g/cm³, whereas the calculated average density (2.47 g/cm³) is slightly higher than the measured average density (2.44 g/cm³). This difference could be due to the contribute of Well 6, which has a wider and a higher range of density values.

4.6. Volturino Unit

Seven wells out of thirteen that penetrated the Apulian Platform carbonates encountered this turbiditic foredeep sequence that locally overlies the Apulian Carbonate Platform Unit. The lack of it in the other wells is either due to tectonic erosion by the overthrusting units or because being deposited in a laterally discontinuous basin.

Sonic logs do not show any detectable velocity increase with depth

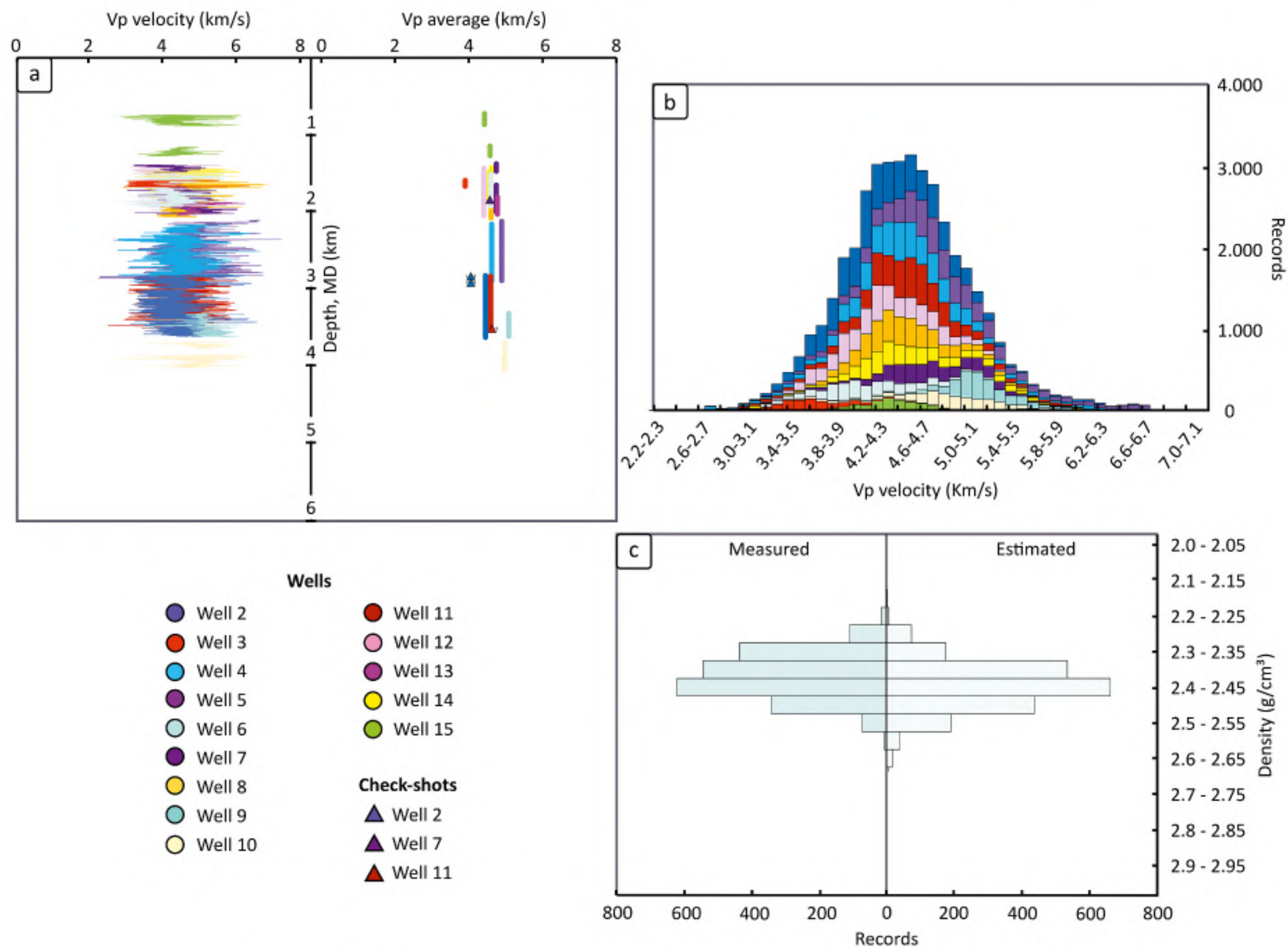


Fig. 10. Calculated densities distribution range of the Lagonegro clastic and carbonatic units.

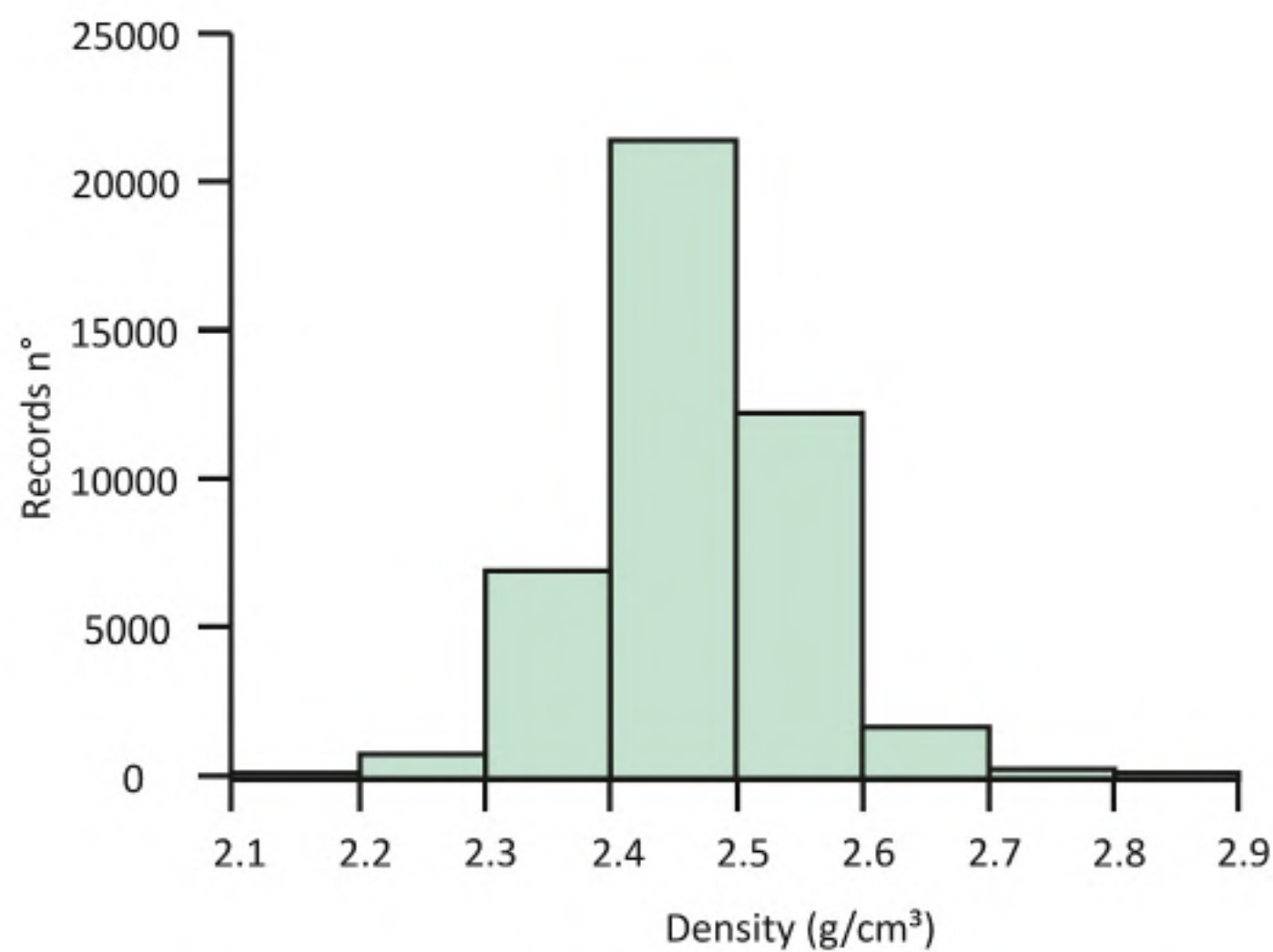


Fig. 11. Velocity and density analysis in the Irpinia 1 unit. (a) Sonic logs and Average Vp discrete logs vs depth cross-plots; (b) Vp distribution histogram; (c) bi-directional bar chart of the measured and calculated densities.

(Fig. 14a). For this unit, only one check-shot measurement is available (triangle in Fig. 14a), being in good agreement with the velocity values from sonic logs.

The histogram (Fig. 14b) shows a wide velocity range, generally unimodal with a mode at 4300–4600 m/s, with a slightly right-skewed distribution. The mean velocity value ranges between 5035 m/s (Well 6) and 3813 m/s (Well 14); the average velocity is 4349 m/s.

The shallowest occurrence of the unit is at 2175 m MD (Well 15),

while the most profound is at 3863 m MD (Well 12). The penetrated total thickness ranges between 191 m (Well 14) and 537 m (Well 8).

The measured density is available for one well 14. The comparison (Fig. 14c) shows that the mode of the measured density is at 2.50–2.60 g/cm³, with an average density of 2.5 g/cm³, while the mode of the calculated density is at 2.50–2.55 g/cm³, with an average density of 2.53 g/cm³. Thus, also in this case, the calculation of synthetic density logs for the rest of the wells can be considered reliable.

4.7. Apulian Carbonate Platform Unit

Among the thirteen wells which penetrate this unit, the sonic logs and discrete Vp do not show any detectable velocity increasing with depth (Fig. 15a). A similar trend is shown by the check-shots interval velocities (triangles in Fig. 15a). These observations agree with sonic logs recorded using similar methodologies on the massive limestones of the Northern Apennines (Trippetta et al., 2021).

The histogram of velocity distribution (Fig. 15b) shows that recorded velocities are characterized by a narrow velocity range aligned into a cumulative Gaussian unimodal distribution with a mode at 6100–6400 m/s. The mean velocity values recorded in the single wells range between 6438 m/s (Well 10) and 5821 m/s (Well 7); the average velocity of the entire unit is 6141 m/s.

The shallowest occurrence of the unit is at 2512 m MD (Well 15 = , while the deepest drilled depth is at 5469 m MD (Well 9). The total penetrated thickness ranges between 123 m (Well 5) and 1152 m (Well 15).

The measured density logs are present in 10 wells, and the comparison with calculated synthetic density logs is illustrated in the bi-directional bar chart (Fig. 15c). The comparison shows a similar distribution of density values, with a mode at 2.65–2.75 g/cm³. The average density value is also similar, being 2.67 g/cm³, for measured density and

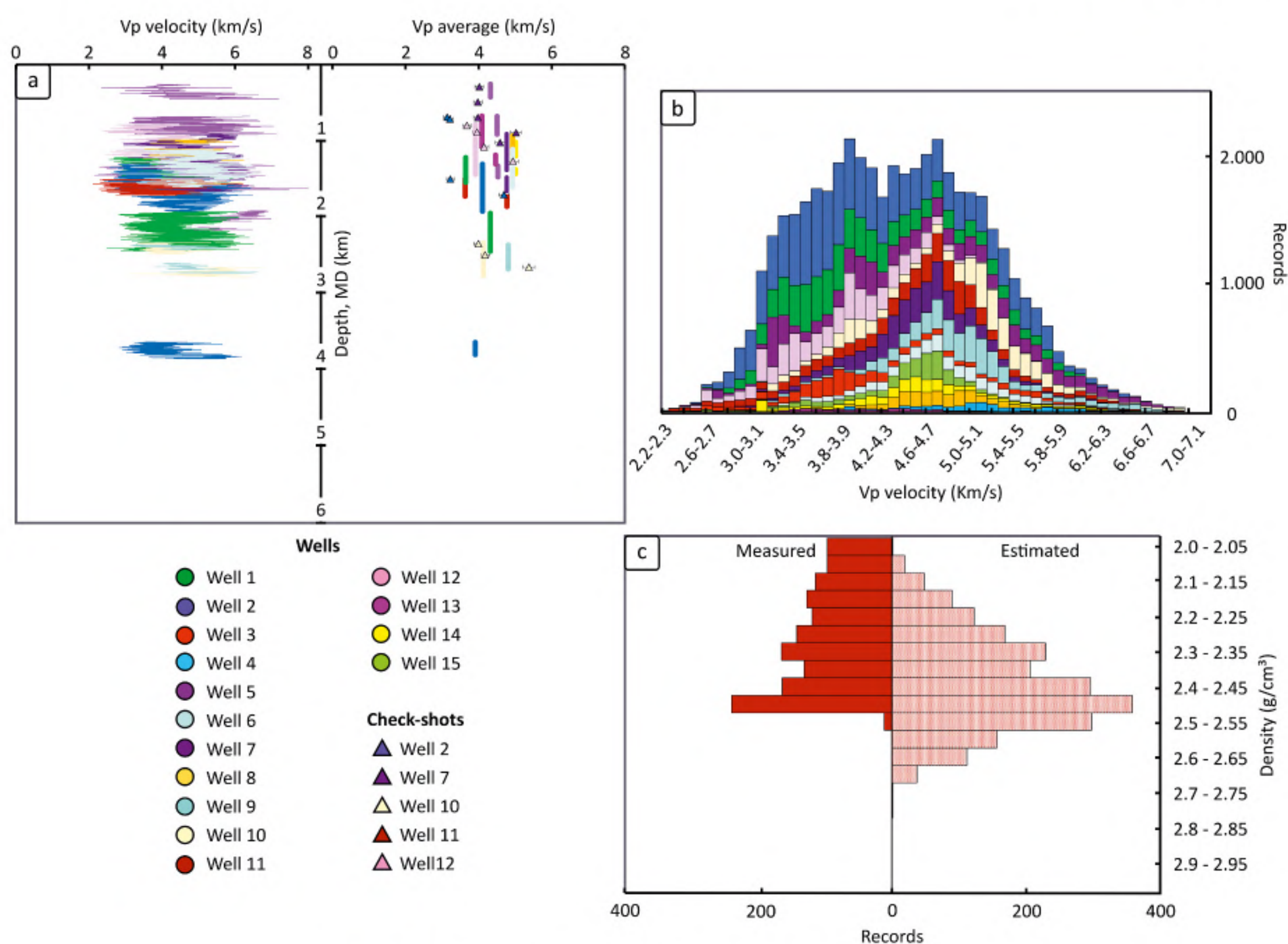


Fig. 12. Calculated densities distribution range of the Irpinia 1 unit.

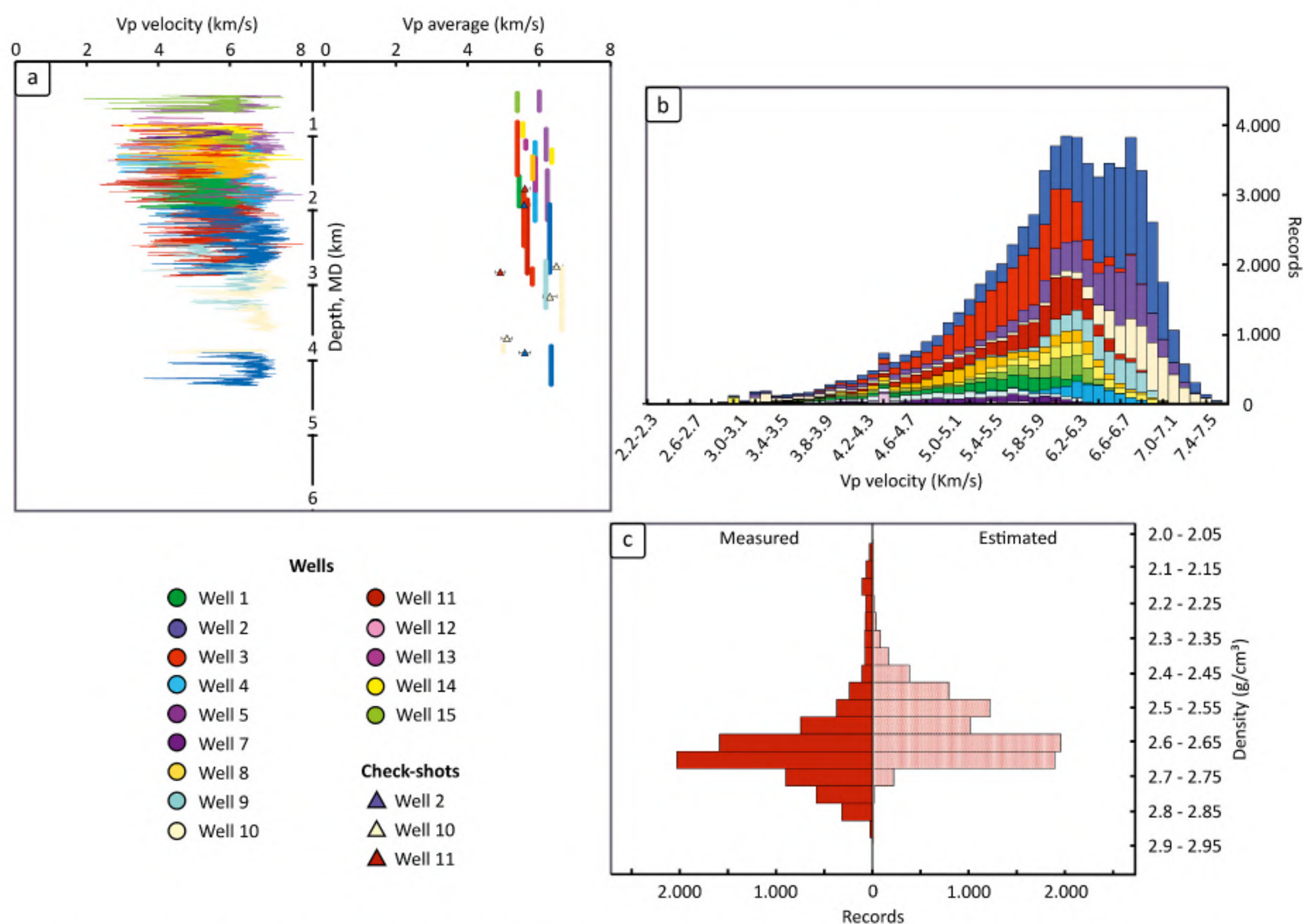


Fig. 13. Velocity and density analysis in the Irpinia 2 and 3 units. (a) Sonic logs and Average Vp discrete logs vs depth cross-plot; (b) Vp distribution histogram; (c) bi-directional bar chart of the measured and calculated densities.

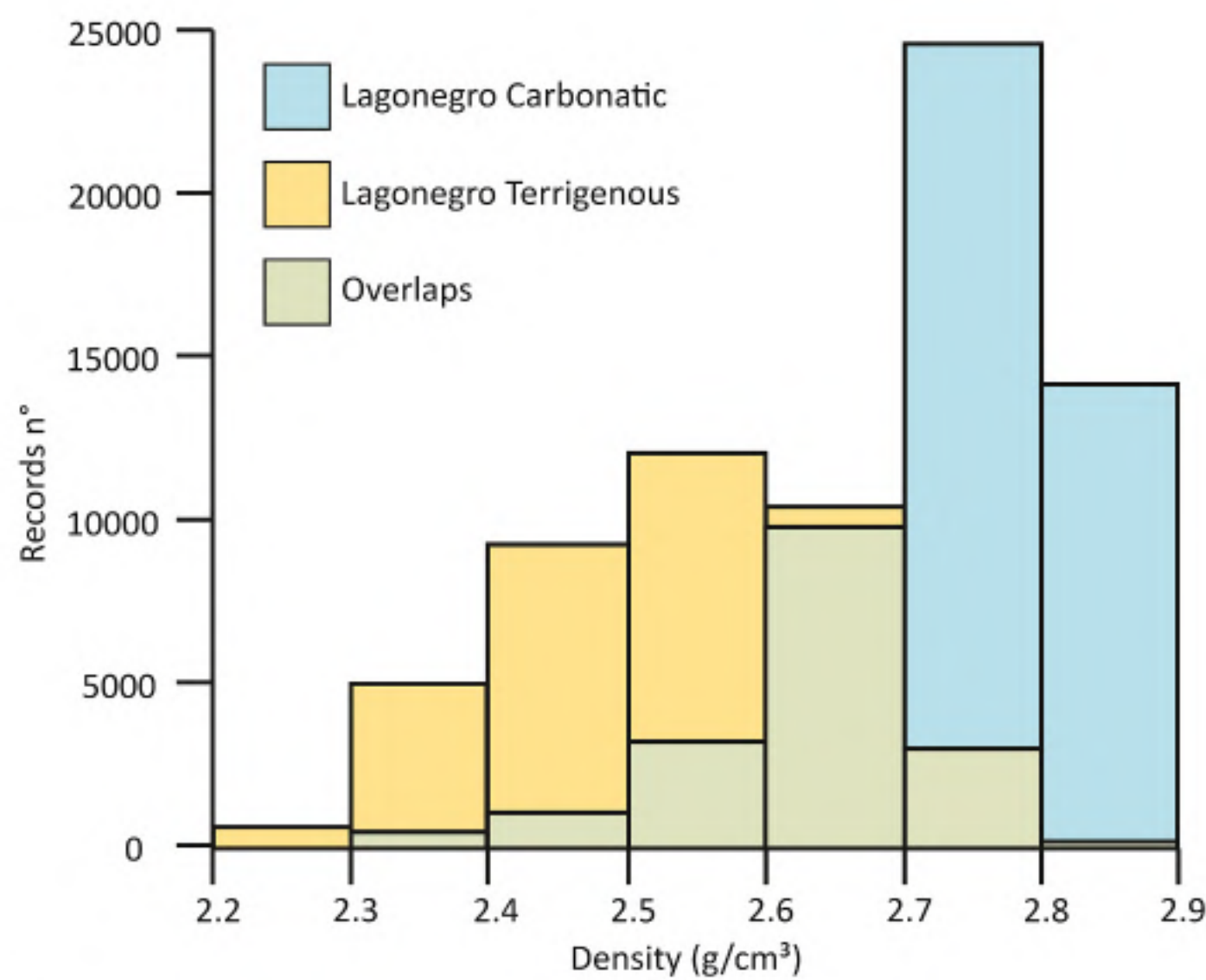


Fig. 14. Velocity and density analysis in the Volturino unit. (a) sonic logs and average Vp discrete logs vs depth cross-plot; (b) Vp distribution histogram; (c) bi-directional bar chart of the measured and calculated densities.

2.66 g/cm³ for calculated density. Thus, the calculation of synthetic density logs can be considered reliable.

5. Discussion

5.1. Comparison between sonic log and check shot interval velocities

As we stated in the methodological part, the sonic log provides

continuous measurements of velocity at high frequency (in general every 30 cm), whilst check-shots are acquired at single points along the borehole section, providing depth-specific measurements.

Within each unit and for each well, sonic logs offer a detailed representation of the velocity variations with depth. These small-scale variations can be explained by considering that the sonic log velocities are influenced by even small-scale heterogeneities within the units, comprising extreme values related to, e.g., local effects of intense fracturing or post-depositional diagenesis. Moreover, despite the editing operated during their calibration, sonic curves may still be affected by invasion, hole rugosity and general borehole drilling problems. Beside these local, high-frequency variations, the average value of the sonic log velocity is representative of the overall behaviour of the drilled interval. In contrast, the check-shot interval velocities obtained from a wider rock volume are less influenced by these small-scale heterogeneities.

For the five wells where the check-shot data are available, it is possible to compare the velocity measurements acquired with the two different techniques. A comprehensive synthetic view of the results, already shown in section 4 (Fig. 7a to 15a), is provided in Table 1. The differences between sonic log velocities and check shots are also illustrated in detail in Fig. S2, for the representative case of the Apulian Platform.

Table 1 demonstrates that in most cases, despite the different acquisition techniques, scale and resolution, the check-shot interval velocities nicely correspond to the average sonic log velocity, with minor differences (generally <5%). Moreover, the results in Table 1 do not show any clear tendency of the sonic log velocity values to be faster or slower with respect to the corresponding check-shot velocities. Moreover, the velocity values are in agreements with recent literature data, acquired in similar lithologies (Trippetta et al., 2021).

In few cases, however, check-shots velocity values differ significantly (10% or more) from the average sonic log velocity of the drilled unit.

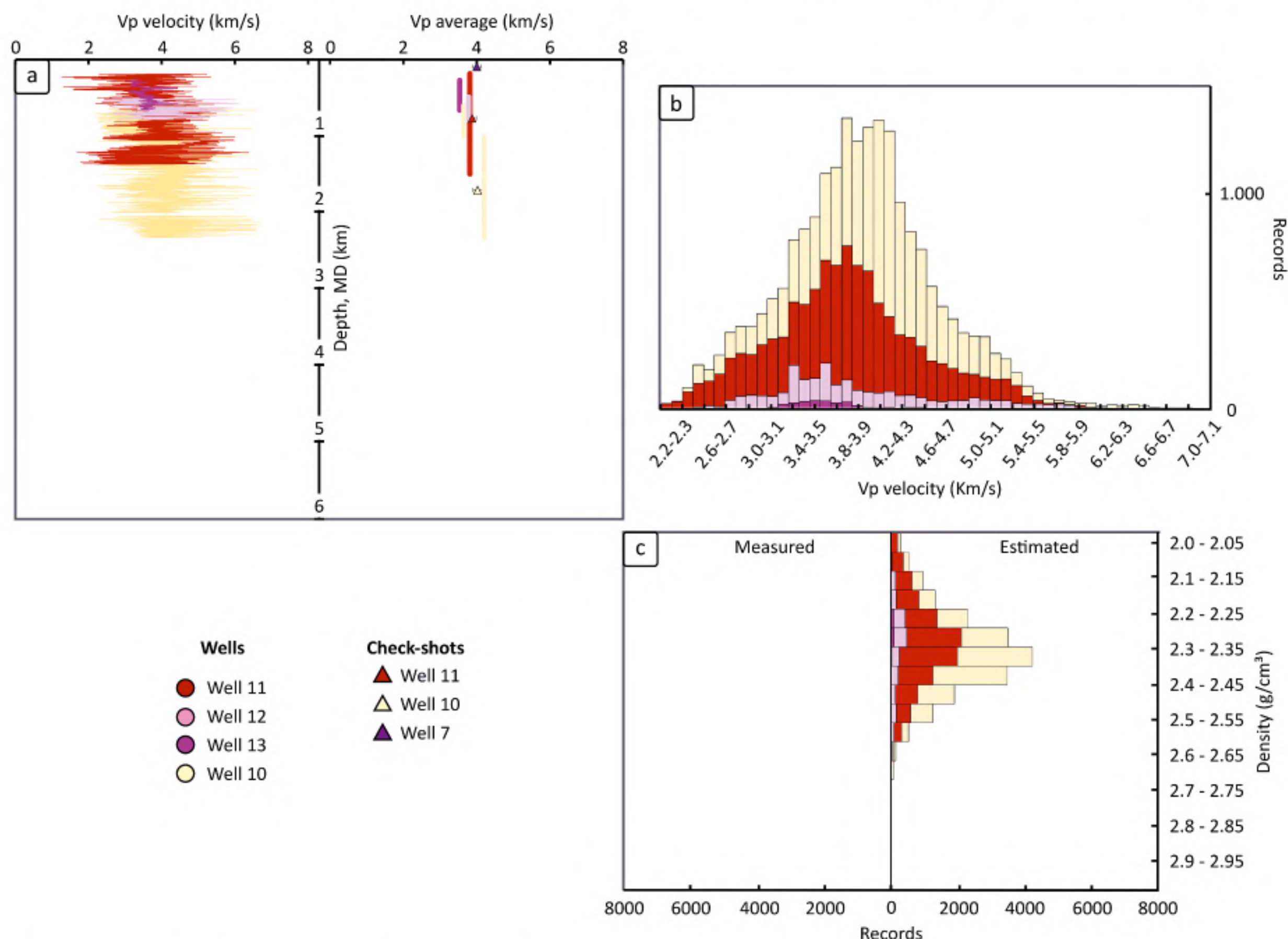


Fig. 15. Velocity and density analysis of the Apulian Platform Carbonates Unit. (a) sonic logs and average Vp discrete logs vs depth cross-plot; (b) Vp distribution histogram; (c) bi-directional bar chart of the measured and calculated densities.

Table 1

Comparison between check shot and sonic log velocities for the five wells where both measurements are available. The percentage differences are indicated in red color, when the sonic log values are higher; in blue color, when the check shot values are higher.

Wells	Check shot Vp, m/s	Sonic Log Vp, m/s	Difference, %
Apulian Platform			
Well 12	6000	5906	1.58
	6235	5906	5.42
Well 7	6000	5821	3.03
	6400	5821	9.48
	5887	5821	1.13
Well 10	6000	6267	-4.35
	6353	6267	1.37
	6400	6267	2.10
	6667	6267	6.19
Volturino			
Well 7	3790	3912	-3.17
	4810	4648	3.43
	4556	4648	-2.00
Well 12	4554	4510	0.98
Irpinia 2 & 3			
Well 12	4105	3990	2.85
Well 7	3790	3613	4.78
Well 10	4000	4101	-2.50
	4167	4101	1.59
	4148	4101	1.14
Irpinia 1			
Well 7	4565	4744	-3.85
Well 10	5072	4970	2.04
Well 2	4051	4448	-9.34
Well 11	4886	4590	6.26

Wells	Check shot Vp, m/s	Sonic Log Vp, m/s	Difference, %
Lagonegro Clastic			
Well 12	3660	3900	-6.35
	3939	3900	1.00
	4132	3900	5.78
Well 7	4565	4664	-2.15
	4565	4652	-1.89
Well 10	4157	4115	1.02
	3973	4115	-3.51
	5072	4970	2.04
Well 2	3213	3359	-4.47
	3574	3891	-8.49
	4674	4742	-1.45
Lagonegro Carbonate			
Well 10	5352	5314	0.72
	6264	6594	-5.13
	6448	6594	-2.24
Well 2	5549	6267	-12.16
	5571	6308	-12.42
Well 7	5000	5174	-3.42
Well 11	5571	5643	-1.28
Liguride			
Well 10	4133	4115	0.44
	3891	4115	-5.60
Well 7	3960	na	na
Well 11	3833	3779	1.42

Slower velocity values of check-shot may be indicative of high fracture density (e.g. fault zones) and/or weaker lithological intervals (e.g. marly/shaly intervals in a calcareous unit). Faster velocity values, in contrast, are generally related to local interlayering of a stronger lithology (e.g., calcareous intervals within the Lagonegro - Clastic units, or a dolostone interval within the carbonate platform units), but may be also indicative of a well-cemented interval.

It can be noted that the largest discrepancies are found within the Lagonegro units, whose large lithological heterogeneity may locally affect the general velocity trend (Fig. 12a and 14a).

5.2. Velocity variations with depth

An overview of the velocity variation with depth for all the considered units and all the wells is offered by the cross-plot of Fig. 16. The same data is reported in Table S1. The distribution with the depth of the values reflects the structural position of each unit within the Southern Apennines tectonic pile, where the Liguride Unit is the shallowest, and the Apulian Carbonate Platform Unit is the deepest.

The overall collected data (Fig. 16, see details in Fig. 7a to 15a) show that the main factor influencing the velocity is the lithology: the carbonate units (i.e. Apennine Carbonate Platform, Lagonegro-Carbonates,

Apulian Carbonate Platform) are significantly faster than the clastic/turbidites ones (Liguride, Lagonegro-Clastics, Irpinia 1, 2 and 3, Volturino).

The shallow Liguride Unit is characterized by low velocities, being composed of a Tertiary flysch succession. Similar velocities are also recorded in the Irpinia 2 and Irpinia 3 units, characterized by a succession of Tertiary foredeep turbidites. Slightly higher average velocities characterize the Irpinia 1 Unit, similar to those of the Lagonegro-Clastic Unit, and to those of the Volturino Unit, being characterized by a similar lithology. The Volturino Unit, in particular, show a wide velocity range. The depositional position of the Volturino Unit is on the slope and this could lead to the random presence of platform-derived materials that could increase the velocity of the unit. The Lagonegro-Carbonate Units are characterized by slightly lower velocities, than the platform carbonate velocities of the Apulian Carbonate Platform Unit (Table 2a). Such observation agrees with the presence of marly intervals in the Lagonegro basinal succession.

Based on both literature and theory (Athy, 1930; Wyllie et al., 1956; Goetz et al., 1979; Mavko et al., 2009; Allen and Allen, 2013), it is expected that mechanical compaction of sedimentary rocks produces more or less pronounced increase of velocity with depth, depending on many different factors, such as lithology, age of the rocks, fracture intensity,

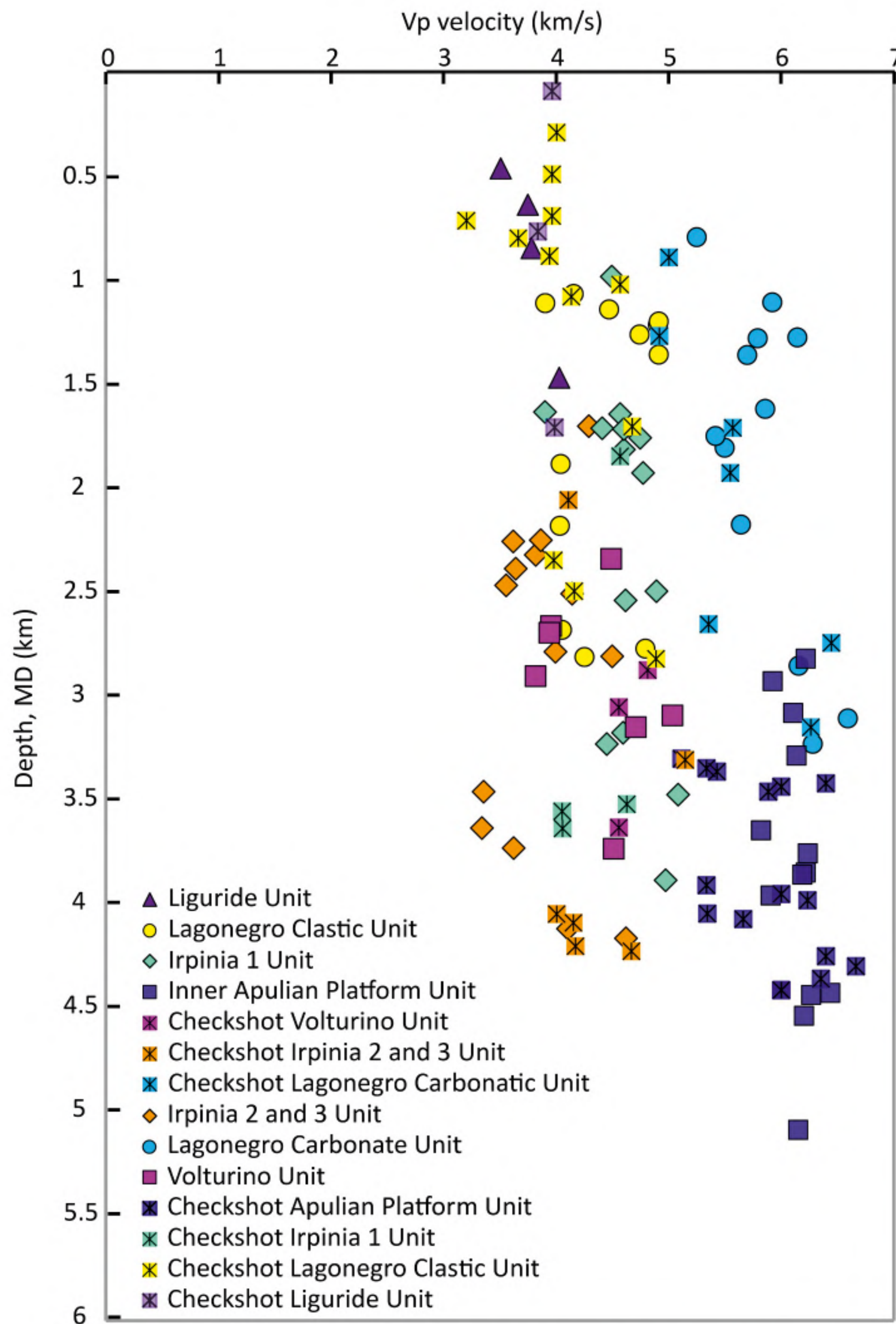


Fig. 16. Summary of the sonic log and check-shot data analyzed in this study and their distribution with depth; each point represents the average sonic log velocity and the average depth of the corresponding drilled interval.

maximum experienced burial, etc. In our data, we can recognize a slight increase of velocity in some clastic units, especially at shallow depths (i. e., Liguride, Lagonegro-Clastic) and also in layered basinal carbonates (i. e., Lagonegro-Carbonates). In the latter case, the velocity increase could be related to the closure of fractures and discontinuities, more than sediment compaction, as also observed by [Trippetta et al. \(2021\)](#) on similar lithology in the Central Apennines. In other units, both clastic (i. e., Irpinia) and carbonates (i.e., Apulian Platform), no significant effect (i.e., velocity increase) of depth is registered.

The not-increasing velocity might be related to the high and fast exhumation of this region ([Butler et al., 2004](#); [Corrado et al., 2002](#); [Mazzoli et al., 2008, 2014](#)), which experienced several kms of uplift and unloading in the last 10 Ma, by combined deep-seated thrusting and

shallow low-angle normal faulting. Consequently, the maximum burial experienced by the successions is much larger than the present-day depth, hampering any further effect of compaction.

5.3. Density values analysis

As shown by the data analyzed in section 4 ([Fig. 7c-15c](#)), in general the density values calculated by applying Gardner's empirical formula to the sonic logs for each lithological unit are in good agreement with the values measured in situ. As explained in section 3.3, a further improvement of the calculated values, and a better correspondence with the measured values, was obtained by applying optimized Gardner's coefficients to carbonates and clastic/turbidites units, respectively,

Table 2

Summary of the average measured and calculated densities of the studied litho-structural units. (*) measured values; (**) calculated values.

(a) Average Vp velocity			(b) Average Density		
Clastic	Carbonate		Clastic	Carbonate	
	Basin	Platform		Basin	Platform
Liguride 3763 m/s	Lagonegro 5855 m/s	Apenninic 6000 m/s	Liguride 2.36 g/cm ³ **	Lagonegro 2.66g/cm ³ * 2.63g/cm ³ **	Apenninic n/a
Lagonegro 4429 m/s			Lagonegro 2.35 g/cm ³ * 2.40 g/cm ³ **		
Irpinia 1 4619 m/s			Irpinia 1 2.40g/cm ³ * 2.47g/cm ³ **		
Irpinia 2 + Irpinia 3 3880 m/s			Irpinia 2 + Irpinia 3 2.44g/cm ³ * 2.47g/cm ³ **		
Volturino 4349 m/s		Apulian 6141 m/s	Volturino 2.60g/cm ³ * 2.57g/cm ³ **		Apulian 2.68g/cm ³ * 2.70g/cm ³ **

instead of the standard (default) values, suggested and extensively applied in literature, also in other sectors of the Apennines (e.g. [Montone and Mariucci, 2020](#)).

[Fig. 17](#) shows the distribution histograms for measured density ([Fig. 17a](#)), calculated density using default Gardner parameters ([Fig. 17b](#)) and calculated density using adopted parameters ([Fig. 17c](#)) for the Apulian carbonate platform. The comparison highlights that the use of the default parameters resulted in a shift towards the higher values of densities, whilst the use of ad-hoc parameters shows density distribution within the range of measured values.

[Table 2b](#) synthetically reports the average measured and calculated density values derived for each considered unit, showing that the maximum difference between measured and calculated densities is 0.07 g/cm³ (less than 3%) for the Irpinia 1 Unit. As expected, the density values measured/calculated for the clastic/turbidite units (2.35 – 2.47 g/cm³) are significantly lower than that of the carbonate units (2.63 – 2.70 g/cm³). An intermediate value of 2.57–2.60 g/cm³ is observed for the Volturino Unit, possibly due to a peculiar and/or more heterogeneous lithology and/or to a larger structural depth. In fact, [Table 2](#) also shows that the density of the clastic units tendentially increases for deeper units, from 2.36 g/cm³ for the Ligurides to 2.47 g/cm³ for the Irpinia 2 and 3 units.

We are conscious that our approach, may introduce small deviations in density values. We acknowledge that cross-plotting density versus velocity and defining their relationship through a power curve (as proposed e.g. by [Ojha and Sain, 2014](#)) might yield more precise results, especially when a larger amount of measured data is available.

5.4. Comparison with previous studies

In this section, we compare the average velocity ([Table 3](#)) and density ([Table 4](#)) values derived in this study with that reported by previous studies in the same region ([Menardi Noguera and Rea, 2000](#); [Improta et al., 2003](#); [Tiberti et al., 2005](#); [Valoroso et al., 2011](#); [Pajola et al., 2020](#)).

The association between the litho-structural units, composing the upper crust of the study area, and Vp velocities, reported in [Table 3](#), derive from quite different types of datasets and elaborations. The velocity model proposed by [Improta et al. \(2000, 2003\)](#) derives from 2 well sonic logs (Taurasi 1 and Puglia 1) and commercial seismic lines located

in the northern sector of the Southern Apennines (Irpinia region). In turn, [Valoroso et al. \(2011\)](#) proposes hybrid velocity values, derived through a calibration with well velocity logs, laboratory measurements, and results of local earthquake tomography and seismic refraction studies. The velocity values proposed by [Brozzetti \(2011\)](#) are also compiled from different data sources, including the interval velocities of few wells (San Gregorio Magno 1, San Fele 1, Maschito 2, Forenza 1), the already cited velocity model defined for the Northern Irpinia by [Improta et al. \(2000, 2003\)](#) and the data collected by [Bally et al. \(1986\)](#) for the Central Apennines. Finally, the velocity model proposed by [Pajola et al. \(2020\)](#) is the most similar and directly comparable to the results of the present paper, being based on a large number (>50) of well logs, calibrated by vertical seismic profiles (VSP - acquisition technique used during borehole seismic data acquisition).

For the comparison with values from previous studies, the sonic log average velocity values are selected. They agree with the check-shot values and have a continuous dataset for all studied wells and units, while check-shots are present only in some wells and have few measurements in some studied units. The same concept was applied while selecting calculated average density values for comparative analysis.

The most relevant differences are with the model proposed by [Valoroso et al. \(2011\)](#), which generally shows lower values with respect to our model for some of the litho-structural units, the Lagonegro Units in particular. The other considered studies generally agree with our compiled velocity and density values for Tertiary flyschs, foredeep turbidites and Apulian Carbonate Platform Unit, while the values proposed for the Lagonegro Units, show some noticeable discrepancies. It is worth noting that in this study we performed a separate analysis for Lagonegro clastic and carbonate successions, significantly improving the understanding of heterogeneous velocity distribution within the Lagonegro units. Moreover, our sonic logs velocity data, derived from a large number (15) of deep boreholes, is check-shot calibrated.

The previous density dataset ([Table 4](#)) was also derived from different sources and techniques. [Mostardini and Merlini \(1986\)](#) proposed density values resulting from the statistical analysis of a composed dataset, including laboratory measurements of surface samples and wells core samples and values interpreted from log data. Density value ranges proposed by [Menardi Noguera and Rea \(2000\)](#) and [Improta et al. \(2003\)](#) for the different litho-structural units were calibrated by the inversion of a regional gravity data. Finally, [Tiberti et al. \(2005\)](#)

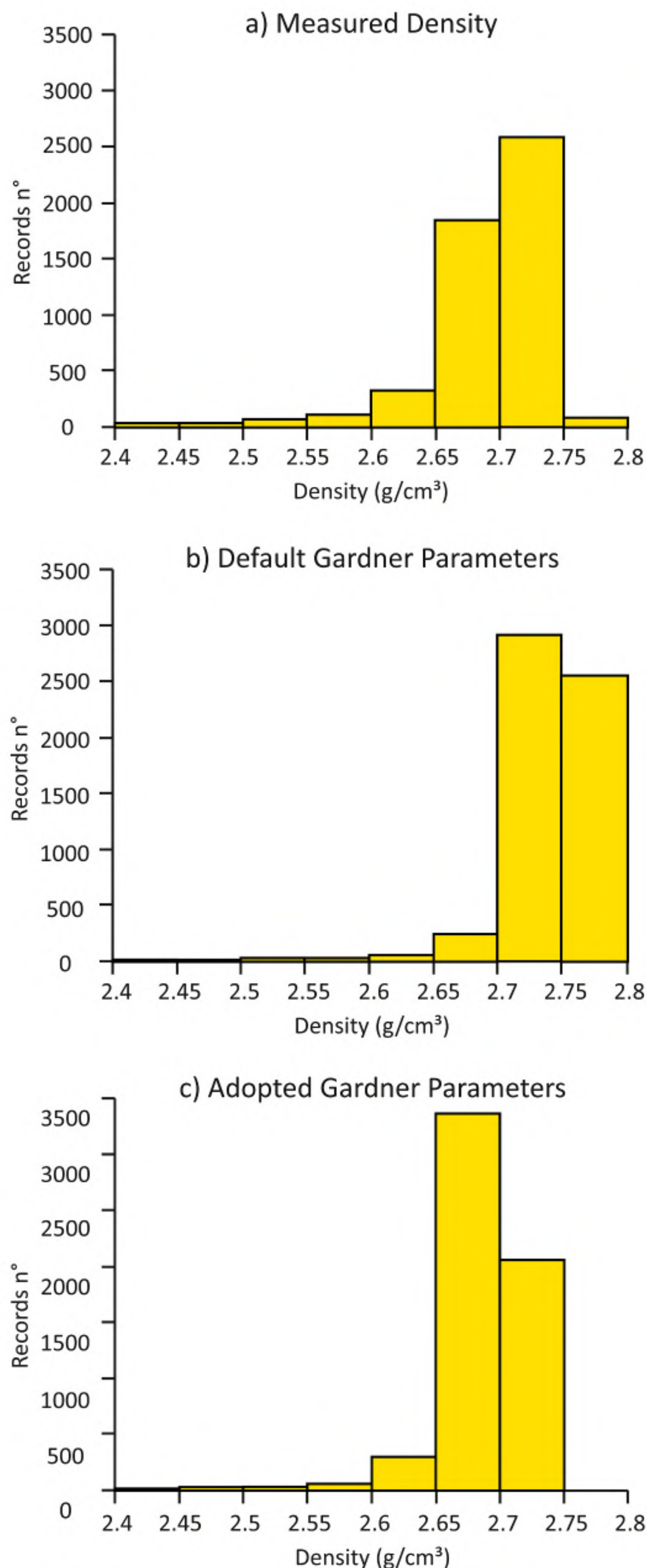


Fig. 17. Comparison of measured density (a) with calculated density log using default Gardner parameters (b) and adopted parameters (c).

compiled and averaged density data from several previously published papers (Carrozzo and Nicolich, 1977; Mostardini and Merlini, 1986; Luongo et al., 1988; Cassinis et al., 1991; Gualteri et al., 1992; Berrino et al., 1998; Gualteri and Zappone, 1998).

It should be noted that in our litho-structural model, we introduced the Volturino Unit, and split the Lagonegro Unit into two lithologically differentiated sub-units (Clastic and Carbonate, respectively), which are not considered in most of the previous studies. Despite these differences, Table 4 shows a generally good correspondence between the results of the present study and the density values derived from the literature, confirming that the clastic/turbidites units are generally less dense than the carbonate units. For the Lagonegro Unit, previous studies indicate a wide range of density values (2.40 g/cm³ to 2.60 g/cm³), reflecting the lithological complexity of this unit.

6. Conclusions

In this paper, we analyze velocity and density data derived from the well-logging of a quite large dataset of deep boreholes located in a relatively small, structurally complex area of the Southern Apennines of Italy.

Considering the obtained velocity data, our results demonstrate a general consistency between velocity values acquired using two different methods, such as sonic logs and check-shots, being in good agreement with recently published results (Trippetta et al., 2021). However, our study shows that the availability of both data types is useful for getting more robust velocity models, particularly in regions characterized by large vertical and horizontal lithological variability.

Considering the density data, we show that simply deriving density from velocity data by applying Gardner's equation with standard coefficient may leave room for variable levels of uncertainties, depending on the different drilled lithologies. Even if limited to a few intervals, calibration with directly acquired density logging can be used for deriving specific Gardner coefficients for each considered lithology/formation, producing improved density data where the direct survey is unavailable.

Finally, here we provide an accurate database of velocities and densities from well logs of the main litho-structural units of the Southern Apennines chain. The average velocity and density values are summarized in Tables 3 and 4, respectively. The deep-sea basinal Lagonegro units – show very wide heterogeneous velocity ranges, reflecting lithological heterogeneity. Our results clearly show that clastic lithologies (Lagonegro-Clastic) have significantly lower velocity and density values, respect to the carbonate ones (Lagonegro-Carbonate).

The obtained velocity and density distribution ranges represent a key input for the future velocity and gravity modellings at both crustal and local scales, as well as for the time to-depth conversion of seismic reflection profiles. The obtained results describe the vertical and lateral lithological heterogeneities between carbonate platforms, deep-sea basinal units and clastic/turbidite successions, in the complex tectonic setting of the Southern Apennines. These values might promote new, improved subsurface geological models, essential for supporting the activities connected to hydrocarbon production in this geologically complex and highly exploited area of the Southern Apennines.

In addition, the study area is a seismically active region covered by an extensive survey of seismic reflection profiles (e.g., Improtta et al., 2000; Patacca and Scandone, 2007a,b; Brozzetti, 2011; Ercoli et al., 2023), thus the new velocity model of the upper crust proposed in this work can also support more accurate earthquake locations and seismogenic faults characterization. Indeed, destructive historical and recent crustal seismic sequences occurred, with maximum magnitude exceeding $M = 7.0$, including e.g., Irpinia 1980, Val d'Agri, 1857 (Pantosti and Valensise, 1990; Benedetti et al., 1998; Boschi et al., 2000; Valoroso et al., 2011; Rovida et al., 2020), where anthropogenic events were also recently registered, related to water impoundment activity (Valoroso et al., 2011; Improtta et al., 2003, 2015).

Even if our analyses referred to a limited, structurally complex area

Table 3

Compilation table of past published studies and present study velocity data.

Litho-structural	Vp (m/s)				
	Present study	Improta et al., 2003	Brozzetti, 2011	Valoroso et al., 2011	Pajola et al., 2020
Liguride Tertiary Flysch	3763	3300-4400	3900	3400-4000	3000-4000
Apennine Platform Carbonates	6000	5300-6000	5400	4800-5500	
Irpinia 1 Foredeep turbidites	4619	4000-5200	4400	3400-4000	3583-4651
Irpinia 2 & 3 Foredeep turbidites	3880				
Lagonegro clastic units	4429	4400-6200	4900	3400-3700	5000-6000
Lagonegro pelagic carbonate units	5855				
Volturino foredeep turbidite unit	4349	n/a	n/a	n/a	n/a
Apulian Platform Carbonates	6141	6000-6500	6000	5900-6200	6294

Table 4

Compilation table of density data derived from past published studies and from the present study.

Litho-structural	Density (g/cm ³)				
	Present study	Mostardini & Merlini, 1986	Tiberti et al., 2005	Improta et al., 2003	Menardi Noguera & Rea, 2000
Liguride Tertiary flysch	2.36	2.45-2.50	n/a	2.45	2.40
Apennine Platform Carbonates	n/a	2.65-2.75	2.67	2.60 - 2.65	n/a
Irpinia 1 Foredeep turbidites	2.47	2.40-2.45	2.40	2.45 - 2.50	2.40
Irpinia 2 & 3 Foredeep turbidites	2.44				
Lagonegro clastic units	2.4	2.50-2.65	2.40-2.55	2.50-2.60	2.40-2.55
Lagonegro basin carbonates units	2.63	2.65-2.70			
Volturino foredeep turbidite unit	2.57	2.65-2.70	n/a	n/a	n/a
Apulian Platform Carbonates	2.7	2.70-2.80	2.67	2.70	2.72

of the Southern Apennines of Italy, the analytical and graphical analysis methods proposed in our study can be readily applied to other case studies worldwide, where similar data sets acquired by well logging are available, in order of building-up a robust and comprehensive model of the subsurface properties of the upper crust. On the other hand, the units drilled by the considered wells, consisting of a platform and basinal carbonates, deposited in the passive continental margin, followed by siliciclastic successions, infilling a migrating syntectonic foredeep, are lithologically comparable to many other fold-and-thrust belts, formed in the peripheral part of a collisional belt.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgements

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.margeo.2024.106634>.